Math 106L In Class 3 Practice Solutions

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Improper Integrals

For all these, correct notation involving limits is essential. I strongly suggest using the sidebar format we did on the corresponding worksheets. When doing substitutions, be sure to change bounds!

1. Compute $\displaystyle\int_1^\infty \frac{1}{x^3}\,dx$.

$$\int_1^\infty \frac{1}{x^3}\,dx = \lim_{b\to\infty}\int_1^b x^{-3}\,dx$$

Sidebar to compute the integral:

$$\int_1^b x^{-3}\,dx = -\frac{1}{2x^2}\bigg|_1^b = -\frac{1}{2b^2} - \left(-\frac{1}{2(1)^2}\right) = -\frac{1}{2b^2} + \frac{1}{2}$$

So

$$\int_1^\infty \frac{1}{x^3}\,dx = \lim_{b\to\infty}\left(-\frac{1}{2b^2} + \frac{1}{2}\right) = 0 + \frac{1}{2} = \frac{1}{2}$$

The integral converges.

2. Compute $\displaystyle\int_1^\infty \frac{1}{\sqrt{x}}\,dx$.

$$\int_1^\infty x^{-1/2}\,dx = \lim_{b\to\infty}\int_1^b x^{-1/2}\,dx$$

Sidebar to compute the integral:

$$\int_1^b x^{-1/2}\,dx = 2x^{1/2}\bigg|_1^b = 2\sqrt{b} - 2\sqrt{1} = 2\sqrt{b} - 2$$

So

$$\int_1^\infty \frac{1}{\sqrt{x}}\,dx = \lim_{b\to\infty}\left(2\sqrt{b} - 2\right) = \infty$$

The integral diverges.

3. Compute $\displaystyle\int_0^\infty e^{-3x}\,dx$.

$$\int_0^\infty e^{-3x}\,dx = \lim_{b\to\infty}\int_0^b e^{-3x}\,dx$$

Sidebar to compute the integral:

$$\int_0^b e^{-3x}\,dx = -\frac{1}{3}e^{-3x}\bigg|_0^b = -\frac{1}{3}e^{-3b} - \left(-\frac{1}{3}e^{0}\right) = -\frac{1}{3}e^{-3b} + \frac{1}{3}$$

So

$$\int_0^\infty e^{-3x}\,dx = \lim_{b\to\infty}\left(-\frac{1}{3}e^{-3b} + \frac{1}{3}\right) = 0 + \frac{1}{3} = \frac{1}{3}$$

The integral converges.

4. Compute $\displaystyle\int_e^\infty \frac{1}{x(\ln x)^2}\,dx$.

$$\int_e^\infty \frac{1}{x(\ln x)^2}\,dx = \lim_{b\to\infty}\int_e^b \frac{1}{x(\ln x)^2}\,dx$$

Sidebar: Substitution to compute the integral

Let $u = \ln x$, then $du = \frac{1}{x}\,dx$. When $x = e$, $u = 1$. When $x = b$, $u = \ln b$.

$$\int_e^b \frac{1}{x(\ln x)^2}\,dx = \int_1^{\ln b} u^{-2}\,du = -u^{-1}\Big|_1^{\ln b} = -\frac{1}{\ln b} - \left(-\frac{1}{1}\right) = -\frac{1}{\ln b} + 1$$

So

$$\int_e^\infty \frac{1}{x(\ln x)^2}\,dx = \lim_{b\to\infty}\left(-\frac{1}{\ln b} + 1\right) = 0 + 1 = 1$$

The integral converges.

5. Compute $\displaystyle\int_{-\infty}^0 e^{2x}\,dx$.

$$\int_{-\infty}^0 e^{2x}\,dx = \lim_{a\to-\infty}\int_a^0 e^{2x}\,dx$$

Sidebar to compute the integral:

$$\int_a^0 e^{2x}\,dx = \frac{1}{2}e^{2x}\Big|_a^0 = \frac{1}{2}e^0 - \frac{1}{2}e^{2a} = \frac{1}{2} - \frac{1}{2}e^{2a}$$

So

$$\int_{-\infty}^0 e^{2x}\,dx = \lim_{a\to-\infty}\left(\frac{1}{2} - \frac{1}{2}e^{2a}\right) = \frac{1}{2} - 0 = \frac{1}{2}$$

The integral converges.

6. Compute $\displaystyle\int_0^\infty \frac{1}{1+x^2}\,dx$.

$$\int_0^\infty \frac{1}{1+x^2}\,dx = \lim_{b\to\infty}\int_0^b \frac{1}{1+x^2}\,dx$$

Sidebar to compute the integral:

$$\int_0^b \frac{1}{1+x^2}\,dx = \arctan(x)\Big|_0^b = \arctan(b) - \arctan(0) = \arctan(b) - 0$$

So

$$\int_0^\infty \frac{1}{1+x^2}\,dx = \lim_{b\to\infty}\big(\arctan(b)\big) = \frac{\pi}{2}$$

The integral converges.

7. Compute $\displaystyle\int_1^\infty \frac{\ln x}{x}\,dx$.

$$\int_1^\infty \frac{\ln x}{x}\,dx = \lim_{b\to\infty}\int_1^b \frac{\ln x}{x}\,dx$$

Sidebar: Substitution to compute the integral

Let $u = \ln x$, then $du = \frac{1}{x}\,dx$. When $x = 1$, $u = 0$. When $x = b$, $u = \ln b$.

$$\int_1^b \frac{\ln x}{x}\,dx = \int_0^{\ln b} u\,du = \frac{1}{2}u^2\Big|_0^{\ln b} = \frac{1}{2}(\ln b)^2 - 0$$

So

$$\int_1^\infty \frac{\ln x}{x}\,dx = \lim_{b\to\infty}\left(\frac{1}{2}(\ln b)^2\right) = \infty$$

The integral diverges.

8. Compute $\displaystyle\int_0^1 \frac{1}{\sqrt[3]{x}}\,dx$.

The integrand is discontinuous at $x = 0$.

$$\int_0^1 x^{-1/3}\,dx = \lim_{a\to 0^+}\int_a^1 x^{-1/3}\,dx$$

Sidebar to compute the integral:

$$\int_a^1 x^{-1/3}\,dx = \frac{3}{2}x^{2/3}\Big|_a^1 = \frac{3}{2}(1)^{2/3} - \frac{3}{2}a^{2/3} = \frac{3}{2} - \frac{3}{2}a^{2/3}$$

So

$$\int_0^1 \frac{1}{\sqrt[3]{x}}\,dx = \lim_{a\to 0^+}\left(\frac{3}{2} - \frac{3}{2}a^{2/3}\right) = \frac{3}{2} - 0 = \frac{3}{2}$$

The integral converges.

9. Compute $\displaystyle\int_0^4 \frac{1}{(x-4)^2}\,dx$.

The integrand is discontinuous at $x = 4$.

$$\int_0^4 \frac{1}{(x-4)^2}\,dx = \lim_{b\to 4^-}\int_0^b (x-4)^{-2}\,dx$$

Sidebar to compute the integral:

$$\int_0^b (x-4)^{-2}\,dx = -(x-4)^{-1}\Big|_0^b = -\frac{1}{b-4} - \left(-\frac{1}{0-4}\right) = -\frac{1}{b-4} - \frac{1}{4}$$

So

$$\int_0^4 \frac{1}{(x-4)^2}\,dx = \lim_{b\to 4^-}\left(-\frac{1}{b-4} - \frac{1}{4}\right) = \infty$$

The integral diverges.

10. Compute $\displaystyle\int_2^5 \frac{1}{\sqrt{x-2}}\,dx$.

The integrand is discontinuous at $x = 2$.

$$\int_2^5 (x-2)^{-1/2}\,dx = \lim_{a\to 2^+}\int_a^5 (x-2)^{-1/2}\,dx$$

Sidebar to compute the integral:

$$\int_a^5 (x-2)^{-1/2}\,dx = 2(x-2)^{1/2}\Big|_a^5 = 2\sqrt{5-2} - 2\sqrt{a-2} = 2\sqrt{3} - 2\sqrt{a-2}$$

So

$$\int_2^5 \frac{1}{\sqrt{x-2}}\,dx = \lim_{a\to 2^+}\left(2\sqrt{3} - 2\sqrt{a-2}\right) = 2\sqrt{3} - 0 = 2\sqrt{3}$$

The integral converges.

11. Compute $\displaystyle\int_0^1 \ln x\,dx$.

The integrand is undefined at $x = 0$.

$$\int_0^1 \ln x\,dx = \lim_{a\to 0^+}\int_a^1 \ln x\,dx$$

Sidebar: Integration by Parts to compute the integral

Let $u = \ln x$ and $dv = dx$. Then $du = \frac{1}{x}\,dx$ and $v = x$.

$$\int_a^1 \ln x\,dx = x\ln x\Big|_a^1 - \int_a^1 x\left(\frac{1}{x}\right)\,dx = x\ln x\Big|_a^1 - x\Big|_a^1$$
$$= (1\ln 1 - a\ln a) - (1-a) = -a\ln a - 1 + a$$

So

$$\int_0^1 \ln x\,dx = \lim_{a\to 0^+}(-a\ln a - 1 + a)$$

Using L'Hôpital's Rule for $\lim_{a\to 0^+} a\ln a = \lim_{a\to 0^+}\frac{\ln a}{1/a} = \lim_{a\to 0^+}\frac{1/a}{-1/a^2} = \lim_{a\to 0^+}(-a) = 0$.

Thus, $\lim_{a\to 0^+}(-0 - 1 + 0) = -1$.

The integral converges.

12. Compute $\displaystyle\int_{-1}^1 \frac{1}{x^3}\,dx$.

The integrand has an infinite discontinuity at $x = 0$. We must split the integral:

$$\int_{-1}^1 \frac{1}{x^3}\,dx = \int_{-1}^0 \frac{1}{x^3}\,dx + \int_0^1 \frac{1}{x^3}\,dx$$

Examine the first part:

$$\int_{-1}^0 x^{-3}\,dx = \lim_{b\to 0^-}\int_{-1}^b x^{-3}\,dx$$

Sidebar to compute the integral:

$$\int_{-1}^{b} x^{-3}\,dx = -\frac{1}{2x^2}\Big|_{-1}^{b} = -\frac{1}{2b^2} - \left(-\frac{1}{2(-1)^2}\right) = -\frac{1}{2b^2} + \frac{1}{2}$$

So

$$\int_{-1}^{0} x^{-3}\,dx = \lim_{b\to 0^-}\left(-\frac{1}{2b^2} + \frac{1}{2}\right) = -\infty$$

Since one part of the integral diverges, the entire improper integral diverges.

13. Compute $\displaystyle\int_0^{\pi/2} \sec^2 x\,dx$.

The integrand has a vertical asymptote at $x = \pi/2$.

$$\int_0^{\pi/2} \sec^2 x\,dx = \lim_{b\to \pi/2^-}\int_0^{b} \sec^2 x\,dx$$

Sidebar to compute the integral:

$$\int_0^{b} \sec^2 x\,dx = \tan x\Big|_0^{b} = \tan(b) - \tan(0) = \tan(b)$$

So

$$\int_0^{\pi/2} \sec^2 x\,dx = \lim_{b\to \pi/2^-}\big(\tan(b)\big) = \infty$$

The integral diverges.

14. Compute $\displaystyle\int_{-\infty}^{\infty} \frac{x}{(1+x^2)^2}\,dx$.

Since both bounds are infinite, we split the integral at $x = 0$:

$$\int_{-\infty}^{\infty} \frac{x}{(1+x^2)^2}\,dx = \int_{-\infty}^{0} \frac{x}{(1+x^2)^2}\,dx + \int_0^{\infty} \frac{x}{(1+x^2)^2}\,dx$$

$$\int_0^{\infty} \frac{x}{(1+x^2)^2}\,dx = \lim_{b\to\infty}\int_0^{b} \frac{x}{(1+x^2)^2}\,dx$$

Sidebar: Substitution to compute the integral

Let $u = 1 + x^2$, then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$.

$$\int_0^{b} \frac{x}{(1+x^2)^2}\,dx = \int_1^{1+b^2} \frac{1}{2}u^{-2}\,du = -\frac{1}{2u}\Big|_1^{1+b^2} = -\frac{1}{2(1+b^2)} - \left(-\frac{1}{2(1)}\right) = -\frac{1}{2(1+b^2)} + \frac{1}{2}$$

So

$$\lim_{b\to\infty}\left(-\frac{1}{2(1+b^2)} + \frac{1}{2}\right) = \frac{1}{2}$$

By symmetry (or repeating the process for the left integral), $\int_{-\infty}^{0} \frac{x}{(1+x^2)^2}\,dx = -\frac{1}{2}$.

Therefore, the whole integral converges to $-\frac{1}{2} + \frac{1}{2} = 0$.

15. Compute $\displaystyle\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx$.

The integrand is discontinuous at $x = 2$.

$$\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx = \lim_{b\to 2^-}\int_0^b \frac{x}{\sqrt{4-x^2}}\,dx$$

Sidebar: Substitution to compute the integral

Let $u = 4 - x^2$, then $du = -2x\,dx$, implying $x\,dx = -\frac{1}{2}du$. When $x = 0$, $u = 4$. When $x = b$, $u = 4 - b^2$.

$$\int_0^b \frac{x}{\sqrt{4-x^2}}\,dx = \int_4^{4-b^2} -\frac{1}{2}u^{-1/2}\,du = -u^{1/2}\Big|_4^{4-b^2} = -\sqrt{4-b^2} - (-\sqrt{4}) = -\sqrt{4-b^2} + 2$$

So

$$\int_0^2 \frac{x}{\sqrt{4-x^2}}\,dx = \lim_{b\to 2^-}\left(-\sqrt{4-b^2}+2\right) = 0 + 2 = 2$$

The integral converges.

Basic Differential Equations

Solve the following initial value problems.

1. $\dfrac{dy}{dx} = 3x^2 - 4$, with $y(1) = 2$.

$$y(x) = \int (3x^2 - 4)\, dx = x^3 - 4x + C$$

Using the initial condition $y(1) = 2$:

$$2 = (1)^3 - 4(1) + C \implies 2 = -3 + C \implies C = 5$$

The specific solution is $y(x) = x^3 - 4x + 5$.

2. $y' = \sin(x) + 2x$, with $y(0) = 3$.

$$y(x) = \int (\sin(x) + 2x)\, dx = -\cos(x) + x^2 + C$$

Using the initial condition $y(0) = 3$:

$$3 = -\cos(0) + (0)^2 + C \implies 3 = -1 + C \implies C = 4$$

The specific solution is $y(x) = -\cos(x) + x^2 + 4$.

3. $\dfrac{dy}{dx} = e^{3x}$, with $y(0) = 1$.

$$y(x) = \int e^{3x}\, dx = \frac{1}{3}e^{3x} + C$$

Using the initial condition $y(0) = 1$:

$$1 = \frac{1}{3}e^0 + C \implies 1 = \frac{1}{3} + C \implies C = \frac{2}{3}$$

The specific solution is $y(x) = \frac{1}{3}e^{3x} + \frac{2}{3}$.

4. $y' = \sec^2(x)$, with $y\left(\frac{\pi}{4}\right) = 2$.

$$y(x) = \int \sec^2(x)\, dx = \tan(x) + C$$

Using the initial condition $y\left(\frac{\pi}{4}\right) = 2$:

$$2 = \tan\left(\frac{\pi}{4}\right) + C \implies 2 = 1 + C \implies C = 1$$

The specific solution is $y(x) = \tan(x) + 1$.

5. $\dfrac{dy}{dx} = \dfrac{1}{x}$, with $y(e) = 4$ for $x > 0$.

$$y(x) = \int \frac{1}{x}\, dx = \ln|x| + C$$

Since $x > 0$, $y(x) = \ln(x) + C$. Using the initial condition $y(e) = 4$:

$$4 = \ln(e) + C \implies 4 = 1 + C \implies C = 3$$

The specific solution is $y(x) = \ln(x) + 3$.

6. $y' = \dfrac{2x}{x^2+1}$, with $y(0) = 5$.

$$y(x) = \int \frac{2x}{x^2+1}\,dx$$

Using the substitution $u = x^2+1$, $du = 2x\,dx$:

$$y(x) = \int \frac{1}{u}\,du = \ln|u| + C = \ln(x^2+1) + C$$

Using the initial condition $y(0) = 5$:

$$5 = \ln(0+1) + C \implies 5 = \ln(1) + C \implies C = 5$$

The specific solution is $y(x) = \ln(x^2+1) + 5$.

7. $\dfrac{dy}{dx} = x\sqrt{x^2+9}$, with $y(4) = 0$.

$$y(x) = \int x(x^2+9)^{1/2}\,dx$$

Using the substitution $u = x^2+9$, $du = 2x\,dx \implies \frac{1}{2}du = x\,dx$:

$$y(x) = \int \frac{1}{2}u^{1/2}\,du = \frac{1}{2}\left(\frac{2}{3}u^{3/2}\right) + C = \frac{1}{3}(x^2+9)^{3/2} + C$$

Using the initial condition $y(4) = 0$:

$$0 = \frac{1}{3}(16+9)^{3/2} + C = \frac{1}{3}(25)^{3/2} + C = \frac{1}{3}(125) + C \implies C = -\frac{125}{3}$$

The specific solution is $y(x) = \frac{1}{3}(x^2+9)^{3/2} - \frac{125}{3}$.

8. $y' = \dfrac{(\ln x)^2}{x}$, with $y(1) = 2$.

$$y(x) = \int \frac{(\ln x)^2}{x}\,dx$$

Using the substitution $u = \ln x$, $du = \frac{1}{x}\,dx$:

$$y(x) = \int u^2\,du = \frac{1}{3}u^3 + C = \frac{1}{3}(\ln x)^3 + C$$

Using the initial condition $y(1) = 2$:

$$2 = \frac{1}{3}(\ln 1)^3 + C \implies 2 = 0 + C \implies C = 2$$

The specific solution is $y(x) = \frac{1}{3}(\ln x)^3 + 2$.

9. $\dfrac{dy}{dx} = \ln(x)$, with $y(1) = -1$.

$$y(x) = \int \ln(x)\,dx$$

Using integration by parts with $u = \ln x$ and $dv = dx$, we have $du = \frac{1}{x}\,dx$ and $v = x$:

$$y(x) = x\ln x - \int x\left(\frac{1}{x}\right)\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$$

Using the initial condition $y(1) = -1$:

$$-1 = 1\ln(1) - 1 + C \implies -1 = 0 - 1 + C \implies C = 0$$

The specific solution is $y(x) = x\ln x - x$.

10. $y' = x\cos(x)$, with $y(0) = 1$.

$$y(x) = \int x\cos(x)\,dx$$

Using integration by parts with $u = x$ and $dv = \cos(x)\,dx$, we have $du = dx$ and $v = \sin(x)$:

$$y(x) = x\sin(x) - \int \sin(x)\,dx = x\sin(x) + \cos(x) + C$$

Using the initial condition $y(0) = 1$:

$$1 = 0\sin(0) + \cos(0) + C \implies 1 = 0 + 1 + C \implies C = 0$$

The specific solution is $y(x) = x\sin(x) + \cos(x)$.

11. $y'' = 12x^2 - 6x$, with $y(0) = 1$ and $y'(0) = 2$.

First, we integrate $y''(x)$ to find $y'(x)$:

$$y'(x) = \int (12x^2 - 6x)\,dx = 4x^3 - 3x^2 + C_1$$

Using $y'(0) = 2 \implies 2 = 0 - 0 + C_1 \implies C_1 = 2$. Now we have $y'(x) = 4x^3 - 3x^2 + 2$. Integrating again to find $y(x)$:

$$y(x) = \int (4x^3 - 3x^2 + 2)\,dx = x^4 - x^3 + 2x + C_2$$

Using $y(0) = 1 \implies 1 = 0 - 0 + 0 + C_2 \implies C_2 = 1$.

The specific solution is $y(x) = x^4 - x^3 + 2x + 1$.

12. $y'' = e^x$, with $y(0) = 2$ and $y'(0) = 3$.

First, we integrate $y''(x)$ to find $y'(x)$:

$$y'(x) = \int e^x\,dx = e^x + C_1$$

Using $y'(0) = 3 \implies 3 = e^0 + C_1 \implies C_1 = 2$. Now we have $y'(x) = e^x + 2$. Integrating again to find $y(x)$:

$$y(x) = \int (e^x + 2)\,dx = e^x + 2x + C_2$$

Using $y(0) = 2 \implies 2 = e^0 + 2(0) + C_2 \implies 2 = 1 + C_2 \implies C_2 = 1$.

The specific solution is $y(x) = e^x + 2x + 1$.

13. $y'' = -\cos(x)$, with $y(\pi) = 0$ and $y'(\pi) = 1$.

First, we integrate $y''(x)$ to find $y'(x)$:

$$y'(x) = \int -\cos(x)\,dx = -\sin(x) + C_1$$

Using $y'(\pi) = 1 \implies 1 = -\sin(\pi) + C_1 \implies 1 = 0 + C_1 \implies C_1 = 1$. Now we have $y'(x) = 1 - \sin(x)$. Integrating again to find $y(x)$:

$$y(x) = \int (1 - \sin(x))\,dx = x + \cos(x) + C_2$$

Using $y(\pi) = 0 \implies 0 = \pi + \cos(\pi) + C_2 \implies 0 = \pi - 1 + C_2 \implies C_2 = 1 - \pi$.

The specific solution is $y(x) = x + \cos(x) + 1 - \pi$.

14. $y'' = \dfrac{1}{(x+1)^2}$, with $y(0) = 2$ and $y'(0) = 1$.

First, we integrate $y''(x)$ to find $y'(x)$:

$$y'(x) = \int (x+1)^{-2}\,dx = -(x+1)^{-1} + C_1 = -\frac{1}{x+1} + C_1$$

Using $y'(0) = 1 \implies 1 = -1 + C_1 \implies C_1 = 2$. Now we have $y'(x) = 2 - \frac{1}{x+1}$. Integrating again to find $y(x)$:

$$y(x) = \int \left(2 - \frac{1}{x+1}\right)\,dx = 2x - \ln|x+1| + C_2$$

Using $y(0) = 2 \implies 2 = 0 - \ln(1) + C_2 \implies 2 = 0 + C_2 \implies C_2 = 2$.

The specific solution is $y(x) = 2x - \ln(x+1) + 2$.

15. $y'' = xe^x$, with $y(0) = 1$ and $y'(0) = 0$.

First, we integrate $y''(x)$ using integration by parts. Let $u = x$ and $dv = e^x\,dx$, so $du = dx$ and $v = e^x$:

$$y'(x) = \int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C_1$$

Using $y'(0) = 0 \implies 0 = 0 - 1 + C_1 \implies C_1 = 1$. Now we have $y'(x) = xe^x - e^x + 1$. Integrating again to find $y(x)$, and reusing our result for $\int xe^x\,dx$:

$$y(x) = \int (xe^x - e^x + 1)\,dx = (xe^x - e^x) - e^x + x + C_2 = xe^x - 2e^x + x + C_2$$

Using $y(0) = 1 \implies 1 = 0 - 2 + 0 + C_2 \implies C_2 = 3$.

The specific solution is $y(x) = xe^x - 2e^x + x + 3$.

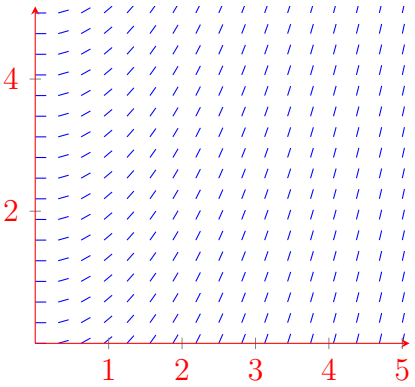
Slopefields and Equilibria

1. Match the following differential equations to their corresponding slope fields. All slope fields are shown in the first quadrant only ($0 \le t \le 5$, $0 \le y \le 5$).

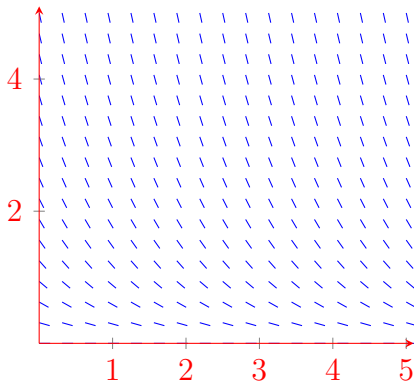
(a) $\frac{dy}{dt} = t - y$ (c) $\frac{dy}{dt} = -y$ (e) $\frac{dy}{dt} = e^{-t}$

(b) $\frac{dy}{dt} = \cos(y)$ (d) $\frac{dy}{dt} = t + y$ (f) $\frac{dy}{dt} = t$

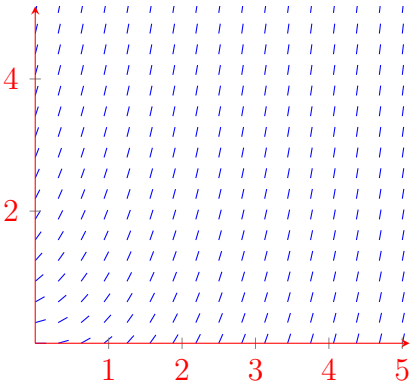
i)



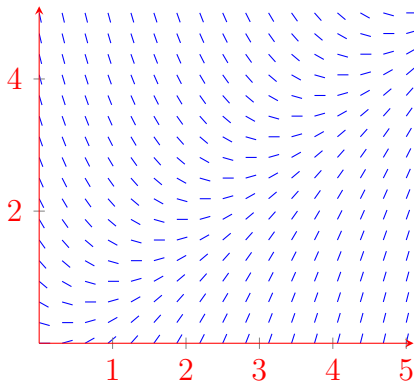
ii)



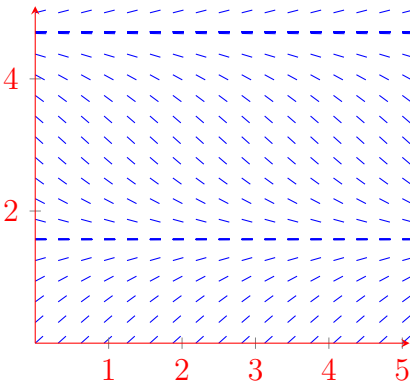
iii)



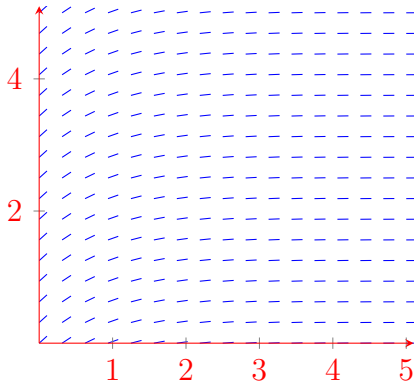
iv)



v)



vi)



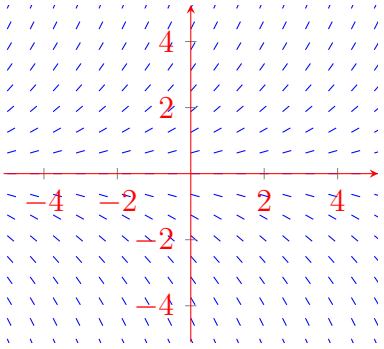
(a) $\frac{dy}{dt} = t - y$ **(iv)** (b) $\frac{dy}{dt} = \cos(y)$ **(v)** (c) $\frac{dy}{dt} = -y$ **(ii)**

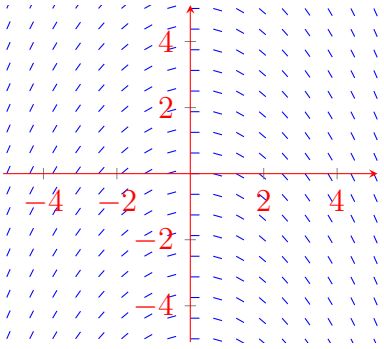
(d) $\frac{dy}{dt} = t + y$ **(iii)** (e) $\frac{dy}{dt} = e^{-t}$ **(vi)** (f) $\frac{dy}{dt} = t$ **(i)**

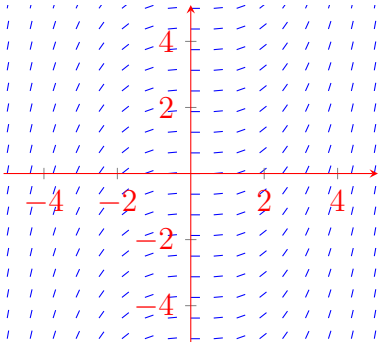
2. Match the following differential equations to their corresponding slope fields. These fields are graphed in all four quadrants ($-5 \le t \le 5$, $-5 \le y \le 5$).

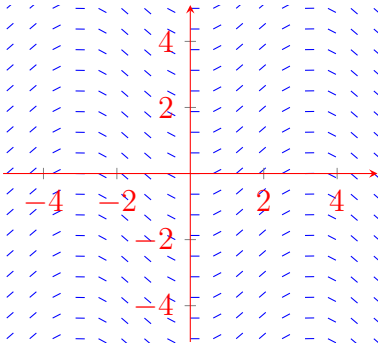
(a) $\dfrac{dy}{dt} = -\dfrac{t}{2}$ (c) $\dfrac{dy}{dt} = \sin(t)$ (e) $\dfrac{dy}{dt} = \dfrac{y}{2}$

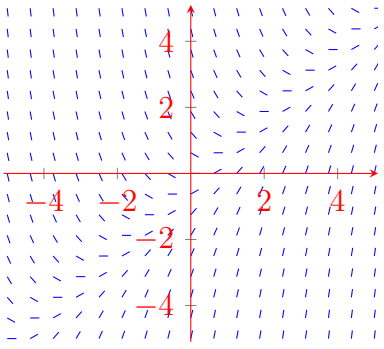
(b) $\dfrac{dy}{dt} = y - t^2$ (d) $\dfrac{dy}{dt} = \dfrac{t^2}{4}$ (f) $\dfrac{dy}{dt} = t - y$

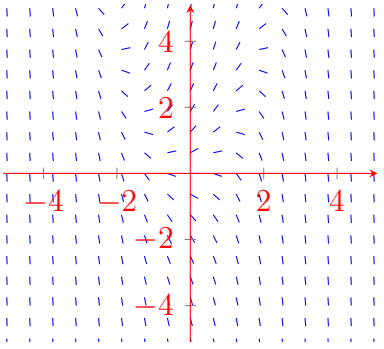
i) 

ii) 

iii) 

iv) 

v) 

vi) 

(a) $\frac{dy}{dt} = -\frac{t}{2}$ **(ii)** (b) $\frac{dy}{dt} = y - t^2$ **(vi)** (c) $\frac{dy}{dt} = \sin(t)$ **(iv)**

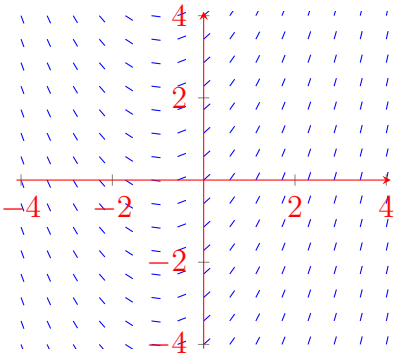
(d) $\frac{dy}{dt} = \frac{t^2}{4}$ **(iii)** (e) $\frac{dy}{dt} = \frac{y}{2}$ **(i)** (f) $\frac{dy}{dt} = t - y$ **(v)**

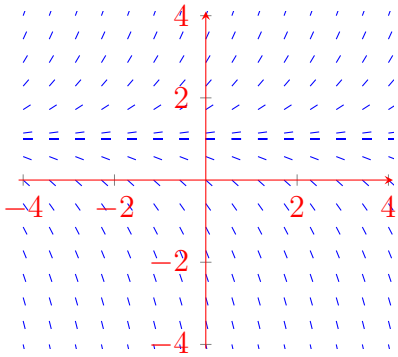
3. Match the differential equations to the slope fields.

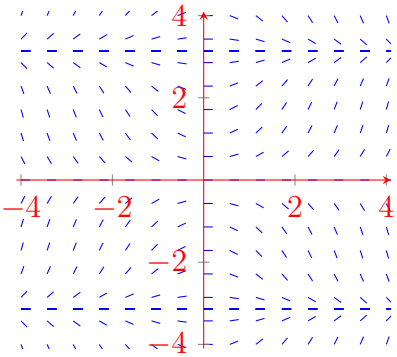
Note: This is not a perfect 1-to-1 matching. Two of the differential equations provided below do not have a corresponding slope field, and two of the slope fields below do not have a corresponding differential equation. If an equation has no match, write "None".

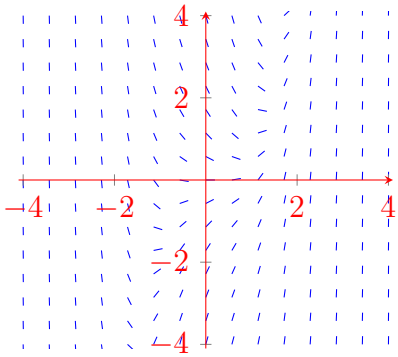
(a) $\frac{dy}{dt} = t^3 - y$ (c) $\frac{dy}{dt} = t^2 y$ (e) $\frac{dy}{dt} = t + 1$

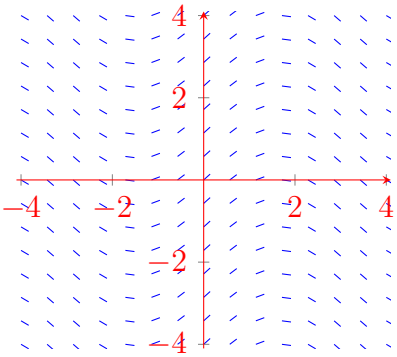
(b) $\frac{dy}{dt} = t\sin(y)$ (d) $\frac{dy}{dt} = y - 1$ (f) $\frac{dy}{dt} = e^t \cos(y)$

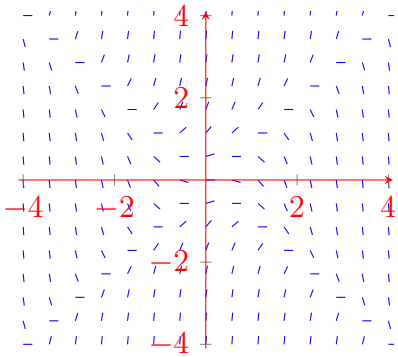
i) 

ii) 

iii) 

iv) 

v) 

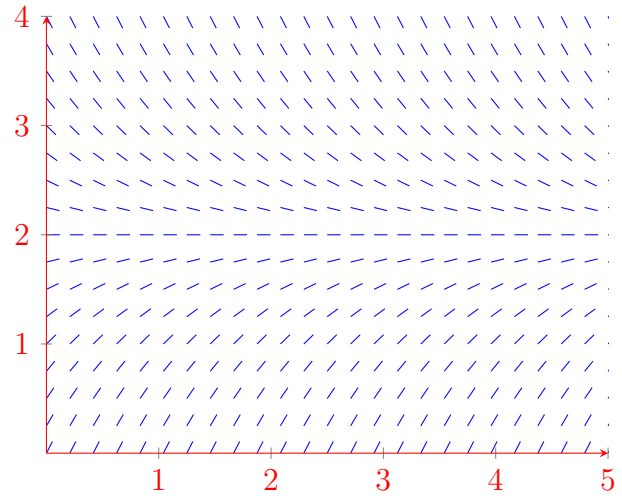
vi) 

(a) $\frac{dy}{dt} = t^3 - y$ **(iv)** (b) $\frac{dy}{dt} = t\sin(y)$ **(iii)** (c) $\frac{dy}{dt} = t^2 y$ **(None)**

(d) $\frac{dy}{dt} = y - 1$ **(ii)** (e) $\frac{dy}{dt} = t + 1$ **(i)** (f) $\frac{dy}{dt} = e^t \cos(y)$ **(None)**

Note: Slope fields (v) and (vi) do not belong to any of the equations given.

4. Find all equilibria of the differential equation $\dfrac{dy}{dt} = 2 - y$. For each equilibrium, determine whether it is stable or unstable.

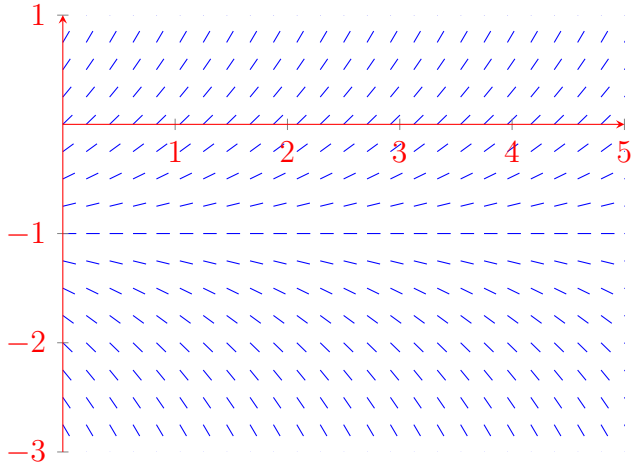


To find the equilibria, we set $\frac{dy}{dt} = 0$:

$$2 - y = 0 \implies y = 2$$

Looking at the slope field or testing values, we see that for $y > 2$, $\frac{dy}{dt} < 0$ (solutions decrease toward 2). For $y < 2$, $\frac{dy}{dt} > 0$ (solutions increase toward 2). Since all nearby solutions tend toward $y = 2$ as $t \to \infty$, $y = 2$ is **stable**.

5. Find all equilibria of the differential equation $\dfrac{dy}{dt} = y + 1$. For each equilibrium, determine whether it is stable or unstable.

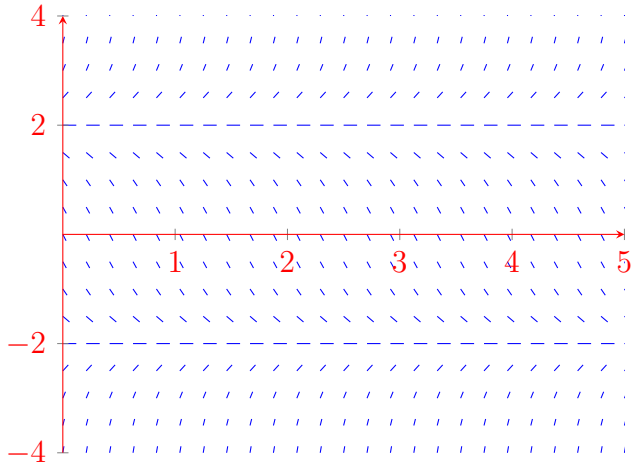


To find the equilibria, we set $\frac{dy}{dt} = 0$:

$$y + 1 = 0 \implies y = -1$$

For $y > -1$, $\frac{dy}{dt} > 0$ (solutions move away from -1). For $y < -1$, $\frac{dy}{dt} < 0$ (solutions also move away from -1). Because nearby solutions do not tend toward the equilibrium, $y = -1$ is **unstable**.

6. Find all equilibria of the differential equation $\frac{dy}{dt} = y^2 - 4$. For each equilibrium, determine whether it is stable or unstable.

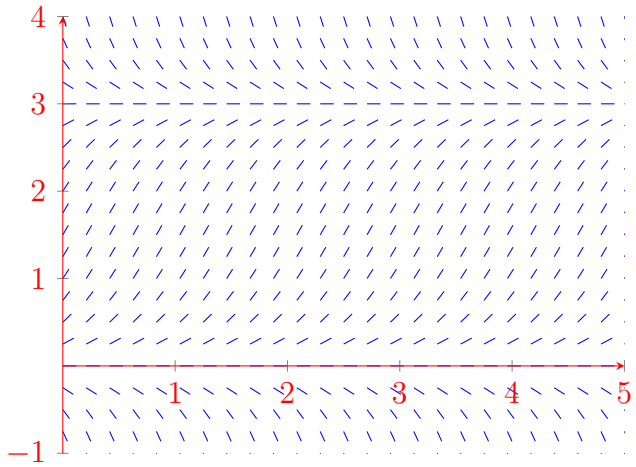


Setting the derivative equal to zero:

$$y^2 - 4 = 0 \implies (y-2)(y+2) = 0 \implies y = 2 \text{ and } y = -2$$

For $y > 2$, $y' > 0$ (moves away). For $-2 < y < 2$, $y' < 0$ (moves away from 2, toward -2). For $y < -2$, $y' > 0$ (moves toward -2). Therefore, $y = 2$ is **unstable** and $y = -2$ is **stable**.

7. Find all equilibria of the differential equation $\frac{dy}{dt} = y(3 - y)$. For each equilibrium, determine whether it is stable or unstable.

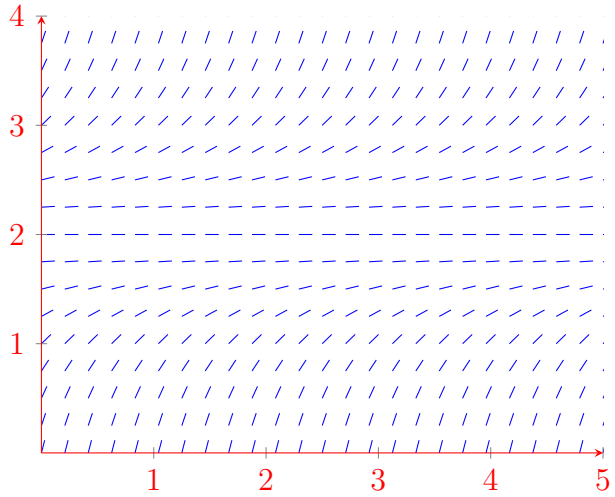


Setting the derivative equal to zero gives:

$$y(3 - y) = 0 \implies y = 0 \text{ and } y = 3$$

For $y > 3$, $y' < 0$ (moves toward 3). For $0 < y < 3$, $y' > 0$ (moves toward 3, away from 0). For $y < 0$, $y' < 0$ (moves away from 0). Therefore, $y = 3$ is **stable** and $y = 0$ is **unstable**.

8. Find all equilibria of the differential equation $\frac{dy}{dt} = (y-2)^2$. For each equilibrium, determine whether it is stable or unstable.

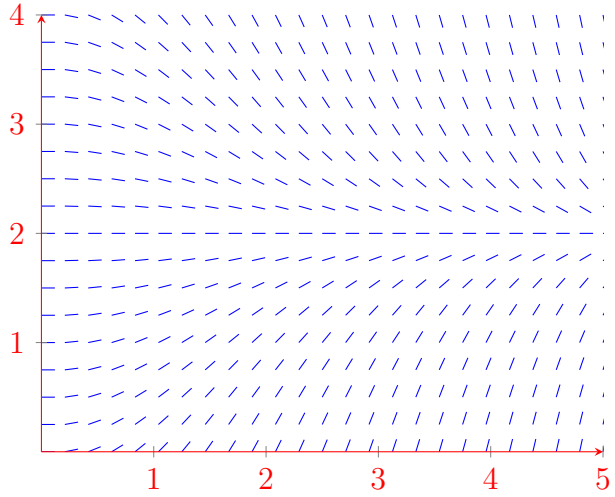


Setting the derivative equal to zero gives:

$$(y-2)^2 = 0 \implies y = 2$$

For $y > 2$, $y' > 0$ (solutions increase and move away from 2). For $y < 2$, $y' > 0$ (solutions increase and move toward 2). Because not *all* nearby solutions tend toward the equilibrium (those above it drift away), $y = 2$ is **unstable**.

9. Find all equilibria of the differential equation $\frac{dy}{dt} = -\frac{t}{2}(y-2)$. For each equilibrium, determine whether it is stable or unstable.

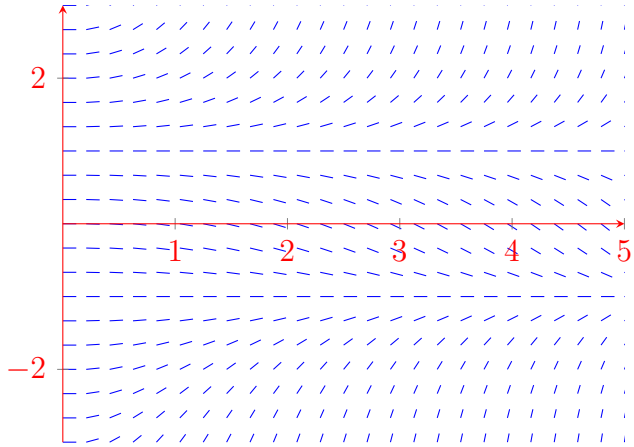


An equilibrium is a constant solution $y(t) = c$ such that $y' = 0$ for all t. Setting the derivative to zero gives:

$$-\frac{t}{2}(y-2) = 0 \text{ for all } t \implies y = 2$$

To determine stability, we look at the behavior as $t \to \infty$. For large positive t, the term $-\frac{t}{2}$ is negative. Therefore, for $y > 2$, $y' < 0$ (solutions move toward 2). For $y < 2$, $y' > 0$ (solutions move toward 2). Since nearby solutions converge to the equilibrium as $t \to \infty$, $y = 2$ is **stable**.

10. Find all equilibria of the differential equation $\frac{dy}{dt} = \frac{t}{4}(y^2 - 1)$. For each equilibrium, determine whether it is stable or unstable.



Setting the derivative equal to zero for all t gives:

$$y^2 - 1 = 0 \implies (y-1)(y+1) = 0 \implies y = 1 \text{ and } y = -1$$

As $t \to \infty$, the coefficient $\frac{t}{4}$ is positive. We check the sign of $y^2 - 1$: For $y > 1$, $y' > 0$ (moves away from 1). For $-1 < y < 1$, $y' < 0$ (moves toward -1, away from 1). For $y < -1$, $y' > 0$ (moves toward -1). Therefore, $y = 1$ is **unstable** and $y = -1$ is **stable**.

Separation of Variables

For each of the following initial value problems, determine if the differential equation is separable. If it is, solve the initial value problem. If it is not, say so.

1. $\dfrac{dy}{dx} = 2xy^2$, with $y(0) = 1$

   This differential equation is separable. We can divide by y^2 and multiply by dx:

   $$\int \frac{1}{y^2}\, dy = \int 2x\, dx$$

   $$-\frac{1}{y} = x^2 + C$$

   Using the initial condition $y(0) = 1$:

   $$-\frac{1}{1} = (0)^2 + C \implies C = -1$$

   Substitute C back and solve for y:

   $$-\frac{1}{y} = x^2 - 1 \implies y = \frac{1}{1 - x^2}$$

2. $\dfrac{dy}{dx} = yxe^{x^2}$, with $y(0) = e$

   This differential equation is separable:

   $$\int \frac{1}{y}\, dy = \int xe^{x^2}\, dx$$

   To integrate the right side, let $u = x^2$. Then $du = 2x\, dx$, which means $\frac{1}{2}du = x\, dx$.

   $$\int xe^{x^2}\, dx = \int e^u \left(\frac{1}{2}\right) du = \frac{1}{2}e^u = \frac{1}{2}e^{x^2}$$

   Setting the integrals equal:

   $$\ln|y| = \frac{1}{2}e^{x^2} + C$$

   Using the initial condition $y(0) = e$:

   $$\ln(e) = \frac{1}{2}e^0 + C \implies 1 = \frac{1}{2} + C \implies C = \frac{1}{2}$$

   Solving for y:

   $$\ln|y| = \frac{1}{2}e^{x^2} + \frac{1}{2} \implies y = e^{\frac{1}{2}e^{x^2} + \frac{1}{2}}$$

3. $\dfrac{dy}{dx} = x + y$, with $y(0) = 1$

   This differential equation is not separable.

4. $\dfrac{dy}{dx} = \dfrac{x^2}{y}$, with $y(1) = 2$

This differential equation is separable:

$$\int y\, dy = \int x^2\, dx$$

$$\frac{1}{2}y^2 = \frac{1}{3}x^3 + C$$

Using the initial condition $y(1) = 2$:

$$\frac{1}{2}(2)^2 = \frac{1}{3}(1)^3 + C \implies 2 = \frac{1}{3} + C \implies C = \frac{5}{3}$$

Solving for y:

$$\frac{1}{2}y^2 = \frac{1}{3}x^3 + \frac{5}{3} \implies y^2 = \frac{2}{3}x^3 + \frac{10}{3}$$

Since $y(1)$ is positive, we take the positive square root:

$$y = \sqrt{\frac{2}{3}x^3 + \frac{10}{3}}$$

5. $\dfrac{dy}{dx} = \dfrac{xe^x}{y^2}$, with $y(0) = 3$

This differential equation is separable:

$$\int y^2\, dy = \int xe^x\, dx$$

To integrate the right side, we use integration by parts. Let $u = x$ and $dv = e^x\, dx$. Then $du = dx$ and $v = e^x$.

$$\int xe^x\, dx = uv - \int v\, du = xe^x - \int e^x\, dx = xe^x - e^x$$

Setting the integrals equal:

$$\frac{1}{3}y^3 = xe^x - e^x + C$$

Using the initial condition $y(0) = 3$:

$$\frac{1}{3}(27) = 0 - e^0 + C \implies 9 = -1 + C \implies C = 10$$

Solving for y:

$$\frac{1}{3}y^3 = xe^x - e^x + 10 \implies y = \sqrt[3]{3xe^x - 3e^x + 30}$$

6. $\dfrac{dy}{dx} = \sin(x + y)$, with $y(0) = \pi$

This differential equation is not separable.

7. $\dfrac{dy}{dx} = xe^{-y}\ln(x)$, with $y(1) = 0$

This differential equation is separable:

$$\int e^y\, dy = \int x\ln(x)\, dx$$

To integrate the right side, we use integration by parts. Let $u = \ln(x)$ and $dv = x\, dx$. Then $du = \frac{1}{x}\, dx$ and $v = \frac{1}{2}x^2$.

$$\int x\ln(x)\, dx = uv - \int v\, du = \frac{1}{2}x^2\ln(x) - \int \left(\frac{1}{2}x^2\right)\left(\frac{1}{x}\right)\, dx$$

$$= \frac{1}{2}x^2\ln(x) - \int \frac{1}{2}x\, dx = \frac{1}{2}x^2\ln(x) - \frac{1}{4}x^2$$

Setting the integrals equal:

$$e^y = \frac{1}{2}x^2\ln(x) - \frac{1}{4}x^2 + C$$

Using the initial condition $y(1) = 0$:

$$e^0 = \frac{1}{2}(1)^2\ln(1) - \frac{1}{4}(1)^2 + C \implies 1 = 0 - \frac{1}{4} + C \implies C = \frac{5}{4}$$

Solving for y:

$$e^y = \frac{1}{2}x^2\ln(x) - \frac{1}{4}x^2 + \frac{5}{4} \implies y = \ln\left(\frac{1}{2}x^2\ln(x) - \frac{1}{4}x^2 + \frac{5}{4}\right)$$

8. $\dfrac{dy}{dx} = \dfrac{x}{y\ln(y)}$, with $y(0) = e$

This differential equation is separable:

$$\int y\ln(y)\, dy = \int x\, dx$$

To integrate the left side, we use integration by parts. Let $u = \ln(y)$ and $dv = y\, dy$. Then $du = \frac{1}{y}\, dy$ and $v = \frac{1}{2}y^2$.

$$\int y\ln(y)\, dy = uv - \int v\, du = \frac{1}{2}y^2\ln(y) - \int \left(\frac{1}{2}y^2\right)\left(\frac{1}{y}\right)\, dy$$

$$= \frac{1}{2}y^2\ln(y) - \int \frac{1}{2}y\, dy = \frac{1}{2}y^2\ln(y) - \frac{1}{4}y^2$$

Setting the integrals equal:

$$\frac{1}{2}y^2\ln(y) - \frac{1}{4}y^2 = \frac{1}{2}x^2 + C$$

Using the initial condition $y(0) = e$:

$$\frac{1}{2}e^2\ln(e) - \frac{1}{4}e^2 = \frac{1}{2}(0)^2 + C \implies \frac{1}{2}e^2(1) - \frac{1}{4}e^2 = C \implies C = \frac{1}{4}e^2$$

Leaving the solution in implicit form:

$$\frac{1}{2}y^2\ln(y) - \frac{1}{4}y^2 = \frac{1}{2}x^2 + \frac{1}{4}e^2$$

9. $\dfrac{dy}{dx} = \dfrac{xy + x}{y}$, with $y(1) = 1$

This differential equation is separable once we factor the numerator to $x(y+1)$:

$$\int \frac{y}{y+1}\, dy = \int x\, dx$$

To integrate the left side, use the substitution $u = y + 1$. Then $du = dy$, and we can solve for y to get $y = u - 1$.

$$\int \frac{y}{y+1}\, dy = \int \frac{u-1}{u}\, du = \int \left(1 - \frac{1}{u}\right)\, du = u - \ln|u|$$

Substituting back $u = y + 1$, the integral is $(y+1) - \ln|y+1|$.

Setting the integrals equal:

$$(y+1) - \ln(y+1) = \frac{1}{2}x^2 + C$$

Using the initial condition $y(1) = 1$:

$$(1+1) - \ln(2) = \frac{1}{2}(1)^2 + C \implies 2 - \ln(2) = \frac{1}{2} + C \implies C = \frac{3}{2} - \ln(2)$$

Leaving the solution in implicit form:

$$y + 1 - \ln(y+1) = \frac{1}{2}x^2 + \frac{3}{2} - \ln(2)$$

Subtracting 1 from both sides gives:

$$y - \ln(y+1) = \frac{1}{2}x^2 + \frac{1}{2} - \ln(2)$$

10. $x\dfrac{dy}{dx} + y = x^2y^2$, with $y(1) = 1$

This differential equation is not separable.

11. $\dfrac{dy}{dx} = e^{x-y}$, with $y(0) = \ln(2)$

This differential equation is separable since $e^{x-y} = e^x e^{-y}$:

$$\int e^y\, dy = \int e^x\, dx$$

$$e^y = e^x + C$$

Using the initial condition $y(0) = \ln(2)$:

$$e^{\ln(2)} = e^0 + C \implies 2 = 1 + C \implies C = 1$$

Solving for y:

$$e^y = e^x + 1 \implies y = \ln(e^x + 1)$$

12. $\dfrac{dy}{dx} = \dfrac{2x}{y(x^2+1)}$, with $y(0) = -2$

This differential equation is separable:

$$\int y\,dy = \int \frac{2x}{x^2+1}\,dx$$

To integrate the right side, let $u = x^2 + 1$. Then $du = 2x\,dx$.

$$\int \frac{2x}{x^2+1}\,dx = \int \frac{1}{u}\,du = \ln|u| = \ln(x^2+1)$$

Setting the integrals equal:

$$\frac{1}{2}y^2 = \ln(x^2+1) + C$$

Using the initial condition $y(0) = -2$:

$$\frac{1}{2}(-2)^2 = \ln((0)^2+1) + C \implies 2 = \ln(1) + C \implies 2 = 0 + C \implies C = 2$$

Solving for y:

$$\frac{1}{2}y^2 = \ln(x^2+1) + 2 \implies y^2 = 2\ln(x^2+1) + 4$$

Since $y(0)$ is negative, we take the negative square root:

$$y = -\sqrt{2\ln(x^2+1)+4}$$

13. $\dfrac{dy}{dx} = \dfrac{1}{x^2 y e^{y^2}}$, with $y(1) = \sqrt{\ln(2)}$

This differential equation is separable:

$$\int ye^{y^2}\,dy = \int x^{-2}\,dx$$

To integrate the left side, let $u = y^2$. Then $du = 2y\,dy$, which implies $\frac{1}{2}du = y\,dy$.

$$\int ye^{y^2}\,dy = \int e^u\left(\frac{1}{2}\right)\,du = \frac{1}{2}e^u = \frac{1}{2}e^{y^2}$$

Setting the integrals equal:

$$\frac{1}{2}e^{y^2} = -x^{-1} + C = -\frac{1}{x} + C$$

Using the initial condition $y(1) = \sqrt{\ln(2)}$:

$$\frac{1}{2}e^{(\sqrt{\ln(2)})^2} = -\frac{1}{1} + C \implies \frac{1}{2}e^{\ln(2)} = -1 + C \implies \frac{1}{2}(2) = -1 + C \implies 1 = -1 + C \implies C = 2$$

Solving for y:

$$\frac{1}{2}e^{y^2} = -\frac{1}{x} + 2 \implies e^{y^2} = 4 - \frac{2}{x} \implies y^2 = \ln\left(4 - \frac{2}{x}\right) \implies y = \sqrt{\ln\left(4 - \frac{2}{x}\right)}$$

14. $\dfrac{dy}{dx} = \sqrt{xy}$, with $y(1) = 4$

This differential equation is separable since $\sqrt{xy} = x^{1/2}y^{1/2}$:

$$\int y^{-1/2}\, dy = \int x^{1/2}\, dx$$

$$2y^{1/2} = \frac{2}{3}x^{3/2} + C$$

Using the initial condition $y(1) = 4$:

$$2(4)^{1/2} = \frac{2}{3}(1)^{3/2} + C \implies 2(2) = \frac{2}{3} + C \implies 4 = \frac{2}{3} + C \implies C = \frac{10}{3}$$

Solving for y:

$$2\sqrt{y} = \frac{2}{3}x^{3/2} + \frac{10}{3} \implies \sqrt{y} = \frac{1}{3}x^{3/2} + \frac{5}{3} \implies y = \left(\frac{1}{3}x^{3/2} + \frac{5}{3}\right)^2$$

15. $\dfrac{dy}{dx} = x\cos(x)(y^2 + 1)$, with $y(0) = 1$

This differential equation is separable:

$$\int \frac{1}{y^2 + 1}\, dy = \int x\cos(x)\, dx$$

To integrate the right side, we use integration by parts. Let $u = x$ and $dv = \cos(x)\, dx$. Then $du = dx$ and $v = \sin(x)$.

$$\int x\cos(x)\, dx = uv - \int v\, du = x\sin(x) - \int \sin(x)\, dx = x\sin(x) - (-\cos(x)) = x\sin(x) + \cos(x)$$

Setting the integrals equal:

$$\arctan(y) = x\sin(x) + \cos(x) + C$$

Using the initial condition $y(0) = 1$:

$$\arctan(1) = (0)\sin(0) + \cos(0) + C \implies \frac{\pi}{4} = 0 + 1 + C \implies C = \frac{\pi}{4} - 1$$

Solving for y:

$$\arctan(y) = x\sin(x) + \cos(x) + \frac{\pi}{4} - 1 \implies y = \tan\left(x\sin(x) + \cos(x) + \frac{\pi}{4} - 1\right)$$

Newton's Law of Cooling

1. Suppose that a bowl of hot soup, which has a temperature of 200°F, is placed in a dining room which is at a constant temperature of 70°F. Suppose that we measure time in minutes after the soup is served, and that the constant of proportionality for its cooling is $k = 0.4$ per minute. By writing down and solving an initial value problem, find how long it will take for the temperature of the soup to cool to a consumable 100°F?

   $\frac{dT}{dt} = -0.4(T - 70)$, $T(0) = 200°$.

   Separate variables: $\int \frac{1}{T-70}\, dT = \int -0.4\, dt$, or $\ln|T - 70| = -0.4t + C$

   Solve this to get $T(t) = Ce^{-0.4t} + 70$, and plug in $200 = Ce^0 + 70$ to find $C = 130$. We get $T(t) = 130e^{-0.4t} + 70$.

   $100 = 130e^{-0.4t} + 70$, which simplifies to $30 = 130e^{-0.4t}$.
   Solving for t gives $t = \frac{\ln(30/130)}{-0.4} \approx 3.67$ minutes.

2. A cold can of soda at 35°F is taken out of a refrigerator and placed on a patio where the ambient temperature is 85°F. The temperature of the soda increases according to Newton's Law of Cooling with a constant of proportionality $k = 0.5$ when time is measured in hours. By writing down and solving an initial value problem, find how long it will take for the soda to warm up to 60°F.

   $\frac{dT}{dt} = -0.5(T - 85)$, $T(0) = 35°$.

   Separate variables: $\int \frac{1}{T-85}\, dT = \int -0.5\, dt$, or $\ln|T - 85| = -0.5t + C$

   Solve this to get $T(t) = Ce^{-0.5t} + 85$, and plug in $35 = Ce^0 + 85$ to find $C = -50$. We get $T(t) = -50e^{-0.5t} + 85$.

   $60 = -50e^{-0.5t} + 85$, which simplifies to $-25 = -50e^{-0.5t}$.
   Solving for t gives $e^{-0.5t} = 0.5 \implies t = \frac{\ln(0.5)}{-0.5} \approx 1.39$ hours.

3. A pizza is removed from a 450°F oven and placed on a counter in a 75°F kitchen to cool. The pizza cools according to a proportionality constant of $k = 0.2$ per minute. Find a formula for the temperature of the pizza as a function of time. How long will it take until the temperature reaches a safe eating temperature of 150°F?

   $\frac{dT}{dt} = -0.2(T - 75)$, $T(0) = 450°$.

   Separate variables: $\int \frac{1}{T-75}\, dT = \int -0.2\, dt$, or $\ln|T - 75| = -0.2t + C$

   Solve this to get $T(t) = Ce^{-0.2t} + 75$, and plug in $450 = Ce^0 + 75$ to find $C = 375$. We get $T(t) = 375e^{-0.2t} + 75$.

   $150 = 375e^{-0.2t} + 75$, which simplifies to $75 = 375e^{-0.2t}$.
   Solving for t gives $e^{-0.2t} = \frac{75}{375} = 0.2 \implies t = \frac{\ln(0.2)}{-0.2} \approx 8.05$ minutes.

4. A piece of glowing steel at 1200°F is plunged into an oil quench bath whose temperature is kept at a constant 100°F. The temperature of the steel decreases according to Newton's Law of Cooling with a constant of proportionality $k = 1.5$ when time is measured in minutes. Find a formula for the temperature of the steel as a function of time. How long will it take until the temperature reaches 200°F?

   $\frac{dT}{dt} = -1.5(T - 100)$, $T(0) = 1200°$.

   Separate variables: $\int \frac{1}{T-100}\, dT = \int -1.5\, dt$, or $\ln|T - 100| = -1.5t + C$

   Solve this to get $T(t) = Ce^{-1.5t} + 100$, and plug in $1200 = Ce^0 + 100$ to find $C = 1100$. We get $T(t) = 1100e^{-1.5t} + 100$.

   $200 = 1100e^{-1.5t} + 100$, which simplifies to $100 = 1100e^{-1.5t}$.
   Solving for t gives $e^{-1.5t} = \frac{100}{1100} \implies t = \frac{\ln(1/11)}{-1.5} \approx 1.60$ minutes.

5. A carton of ice cream is removed from a 10°F freezer and accidentally left on a table in a 72°F room. The rate of temperature change has a proportionality constant of $k = 0.8$ per hour. By writing down and solving an initial value problem, determine how long it will take for the ice cream to reach 30°F (at which point it begins melting rapidly).

   $\frac{dT}{dt} = -0.8(T - 72)$, $T(0) = 10°$.

   Separate variables: $\int \frac{1}{T-72}\, dT = \int -0.8\, dt$, or $\ln|T - 72| = -0.8t + C$

   Solve this to get $T(t) = Ce^{-0.8t} + 72$, and plug in $10 = Ce^0 + 72$ to find $C = -62$. We get $T(t) = -62e^{-0.8t} + 72$.

   $30 = -62e^{-0.8t} + 72$, which simplifies to $-42 = -62e^{-0.8t}$.
   Solving for t gives $e^{-0.8t} = \frac{42}{62} \implies t = \frac{\ln(21/31)}{-0.8} \approx 0.49$ hours (or roughly 29 minutes).

6. A head chef is found dead inside the restaurant's walk-in refrigerator. The refrigerator's temperature is kept at a constant 38°F. The police arrive and measure the temperature of the body at 8:00 AM, finding it to be 68°F. They take a second reading two hours later at 10:00 AM, and the temperature has dropped to 53°F.

   Assume the victim's normal body temperature was 98°F at the exact time of death. Security footage from the hallway shows three employees leaving the walk-in refrigerator that morning:

   - Alice (The Sous Chef) left at 5:00 AM.
   - Bob (The Line Cook) left at 6:00 AM.
   - Charlie (The Manager) left at 7:00 AM.

   By writing and solving a differential equation using Newton's Law of Cooling, determine the exact time of death. Which suspect should the police arrest?

   Let $t = 0$ correspond to 8:00 AM, the time of the first temperature reading.

   $$\frac{dT}{dt} = -k(T - 38)$$

Separating variables and integrating:

$$\int \frac{1}{T-38}\,dT = \int -k\,dt$$

$$\ln|T-38| = -kt + C$$

$$T(t) = Ce^{-kt} + 38$$

Using the condition at 8:00 AM ($t = 0$, $T = 68°$):

$$68 = Ce^{0} + 38 \implies C = 30$$

$$T(t) = 30e^{-kt} + 38$$

Using the condition at 10:00 AM ($t = 2$, $T = 53°$) to find k:

$$53 = 30e^{-2k} + 38 \implies 15 = 30e^{-2k} \implies \frac{1}{2} = e^{-2k}$$

$$\ln(0.5) = -2k \implies k = \frac{\ln(0.5)}{-2} = \frac{\ln(2)}{2} \approx 0.3466 \text{ hr}^{-1}$$

Now, we set $T = 98°$ to find the time of death:

$$98 = 30e^{-kt} + 38 \implies 60 = 30e^{-kt} \implies 2 = e^{-kt}$$

$$\ln(2) = -kt \implies t = -\frac{\ln(2)}{k}$$

Substituting our exact value for k:

$$t = -\frac{\ln(2)}{\left(\frac{\ln(2)}{2}\right)} = -2 \text{ hours}$$

Since $t = 0$ was 8:00 AM, $t = -2$ means the time of death was exactly 2 hours prior, at 6:00 AM. The police should arrest **Bob**.

7. The following four graphs show the temperature $T(t)$ of four different objects cooling over time. All four objects start at an initial temperature of 100°C.

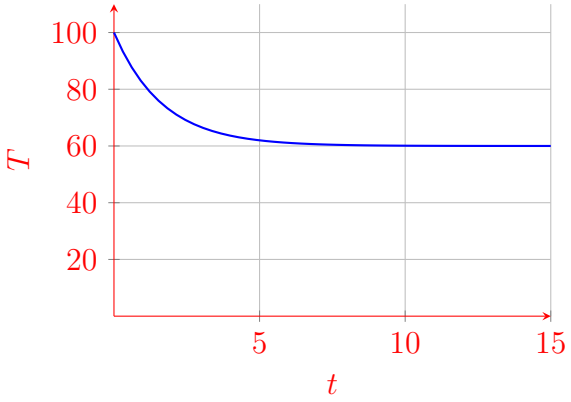
   Match each graph to its corresponding differential equation below by considering the role of the ambient temperature (T^*) and the relative cooling rate (k).

   Note: Two of the differential equations provided do not match any of the graphs. If an equation has no match, write "None".

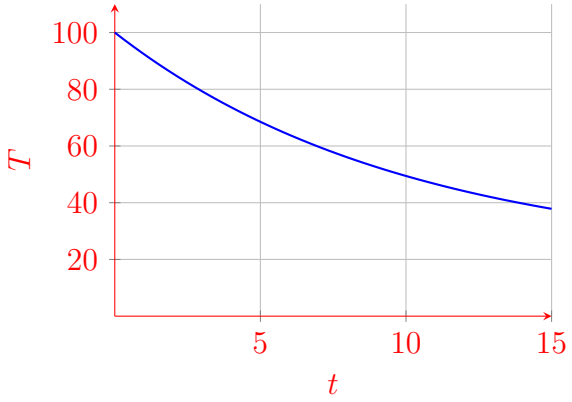
   (a) $\frac{dT}{dt} = -0.6(T - 60)$ (c) $\frac{dT}{dt} = -0.1(T - 60)$ (e) $\frac{dT}{dt} = -0.6(T - 80)$

   (b) $\frac{dT}{dt} = -0.1(T - 20)$ (d) $\frac{dT}{dt} = -0.6(T - 20)$ (f) $\frac{dT}{dt} = -0.1(T - 0)$

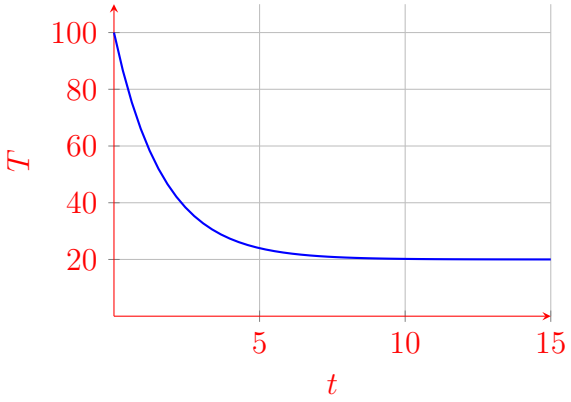
i)



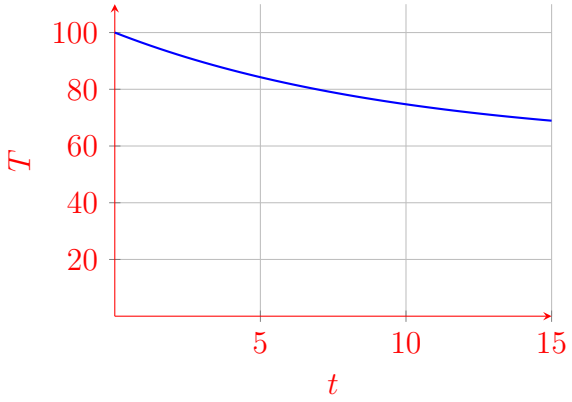
ii)



iii)



iv)



To match the equations, look at two key features of the graphs: the horizontal asymptote (which represents the ambient temperature, T^*) and the steepness of the curve (which represents the cooling rate, k).

- **Graphs (i) and (iv)** level off at 60°C, meaning $T^* = 60$. These must correspond to equations (a) and (c). Graph (i) drops to 60°C much faster than graph (iv), meaning it has the larger decay constant ($k = 0.6$).
- **Graphs (ii) and (iii)** level off at 20°C, meaning $T^* = 20$. These must correspond to equations (b) and (d). Graph (iii) drops to 20°C much faster than graph (ii), meaning it has the larger decay constant ($k = 0.6$).

(a) $\frac{dT}{dt} = -0.6(T - 60)$ **(i)** (c) $\frac{dT}{dt} = -0.1(T - 60)$ **(iv)** (e) $\frac{dT}{dt} = -0.6(T - 80)$ **(None)**

(b) $\frac{dT}{dt} = -0.1(T - 20)$ **(ii)** (d) $\frac{dT}{dt} = -0.6(T - 20)$ **(iii)** (f) $\frac{dT}{dt} = -0.1(T - 0)$ **(None)**

8. Match each physical scenario described below with the differential equation that best models its temperature, $T(t)$.

   When matching, consider the role of the ambient temperature (T^*) and how the material properties (like insulation) affect the relative cooling/heating rate (k).

   Note: Two of the differential equations provided do not match any of the scenarios. If an equation has no match, write "None".

   (a) $\dfrac{dT}{dt} = -0.5(T - 90)$ (c) $\dfrac{dT}{dt} = -0.5(T - 35)$ (e) $\dfrac{dT}{dt} = -0.02(T - 68)$

   (b) $\dfrac{dT}{dt} = -0.5(T - 68)$ (d) $\dfrac{dT}{dt} = -0.02(T - 90)$ (f) $\dfrac{dT}{dt} = -0.02(T - 180)$

   i) A hot mug of coffee (initially 180°F) is poured into a thin, uninsulated paper cup and left on a table in a 68°F living room.

   ii) A hot mug of coffee (initially 180°F) is poured into a highly insulated, double-walled vacuum thermos and left on a table in a 68°F living room.

   iii) A cold soda (initially 35°F) is left in a thin aluminum can (which conducts heat rapidly) on a sunny patio where the temperature is 90°F.

   iv) A cold soda (initially 35°F) is placed inside a thick, heavy-duty foam koozie and left on a sunny patio where the temperature is 90°F.

To match these, we must identify the ambient temperature (T^*) for each location, and determine relative sizes of k. A large k means rapid temperature change (poor insulation), while a small k means slow temperature change (good insulation). Note that the initial temperatures (180°F and 35°F) do not appear in the differential equations themselves, only in the initial conditions.

- **Scenario (i):** The ambient temperature is 68°F, so $T^* = 68$. Because it is in a thin paper cup, it cools rapidly, meaning we want the larger k-value ($k = 0.5$). This matches **(b)**.
- **Scenario (ii):** The ambient temperature is 68°F, so $T^* = 68$. Because it is in an insulated thermos, it cools slowly, meaning we want the smaller k-value ($k = 0.02$). This matches **(e)**.
- **Scenario (iii):** The ambient temperature is 90°F, so $T^* = 90$. Because aluminum conducts heat rapidly, it warms up fast, meaning we want the larger k-value ($k = 0.5$). This matches **(a)**.
- **Scenario (iv):** The ambient temperature is 90°F, so $T^* = 90$. Because the thick foam koozie insulates the drink, it warms up slowly, meaning we want the smaller k-value ($k = 0.02$). This matches **(d)**.

(a) $\frac{dT}{dt} = -0.5(T - 90)$ **(iii)** (c) $\frac{dT}{dt} = -0.5(T - 35)$ **(None)** (e) $\frac{dT}{dt} = -0.02(T - 68)$ **(ii)**

(b) $\frac{dT}{dt} = -0.5(T - 68)$ **(i)** (d) $\frac{dT}{dt} = -0.02(T - 90)$ **(iv)** (f) $\frac{dT}{dt} = -0.02(T - 180)$ **(None)**

Note: Equations (c) and (f) incorrectly use the initial temperature of the object as the ambient temperature.

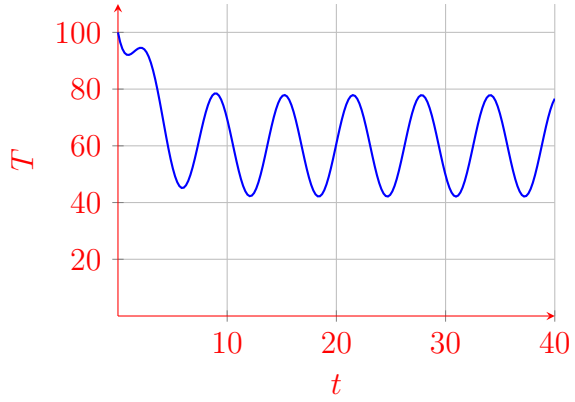
9. The following four graphs show the temperature $T(t)$ of four different objects. Each object starts at an initial temperature of 100°F and is placed in an environment where the ambient temperature fluctuates according to $T^*(t) = 40\sin(t) + 60$.

   Notice that the x-axes have been scaled differently for each graph so that the initial rate of cooling visually appears identical. Match each graph to its corresponding differential equation by observing the **amplitude of the eventual sinusoidal steady-state**.

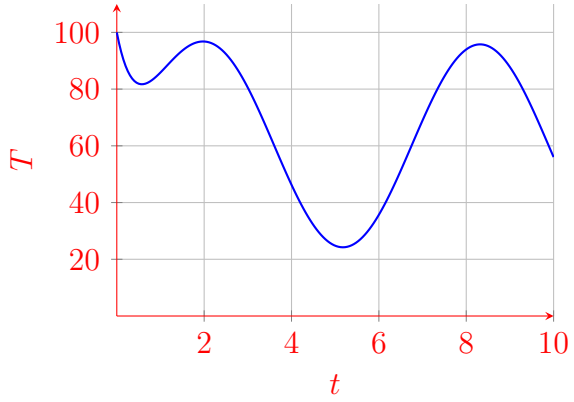
   (a) $\dfrac{dT}{dt} = -2\big(T - (40\sin(t) + 60)\big)$ (c) $\dfrac{dT}{dt} = -0.5\big(T - (40\sin(t) + 60)\big)$

   (b) $\dfrac{dT}{dt} = -1\big(T - (40\sin(t) + 60)\big)$ (d) $\dfrac{dT}{dt} = -0.25\big(T - (40\sin(t) + 60)\big)$

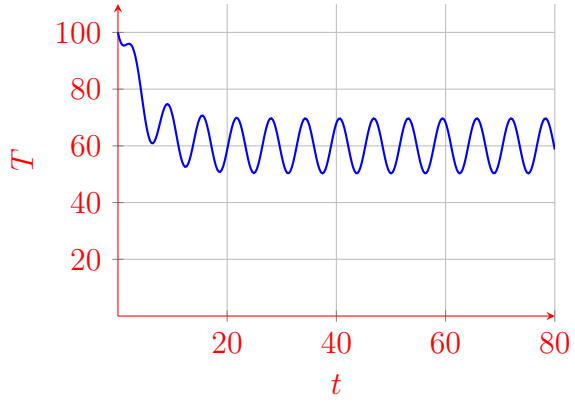
i)



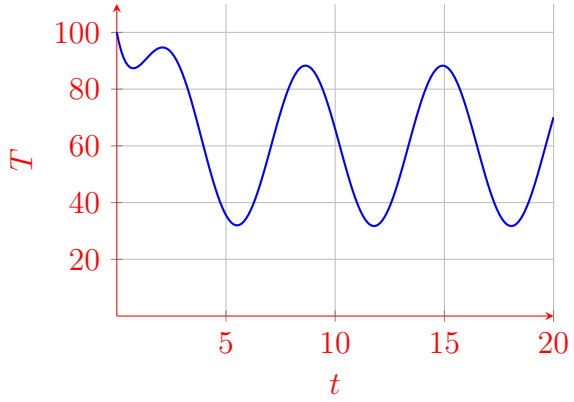
ii)



iii)



iv)



To match these, we consider the physical meaning of k. A large k means the object is poorly insulated and will rapidly track the ambient temperature, resulting in a large steady-state amplitude (close to the ambient amplitude of 40). A small k means the object is highly insulated, so it resists the rapid fluctuations of the ambient temperature, resulting in a very small steady-state amplitude.

- **Graph (ii)** has the largest steady-state amplitude (nearly peaking at 100 and dipping to 20). This corresponds to the largest k, so it matches **(a)** ($k = 2$).
- **Graph (iv)** has the second largest amplitude. This matches **(b)** ($k = 1$).
- **Graph (i)** has a smaller amplitude. This matches **(c)** ($k = 0.5$).
- **Graph (iii)** is highly damped, with the temperature barely oscillating around the 60° baseline. This corresponds to the smallest k, so it matches **(d)** ($k = 0.25$).

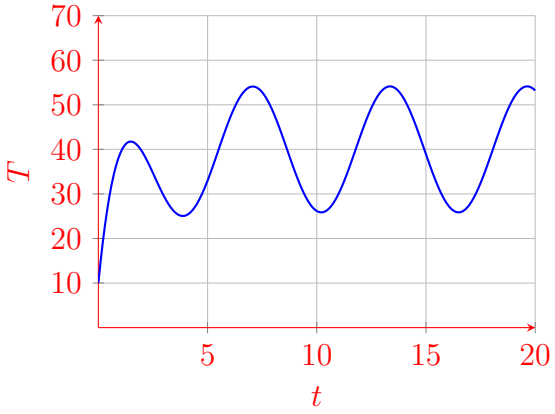
10. The following four graphs show the temperature $T(t)$ of four different objects starting at 10°F, placed into an environment where the temperature fluctuates according to $T^*(t) = 20\cos(t) + 40$.

    Once again, the x-axes have been scaled so the initial warming phase visually matches across all plots. Match the graphs to their differential equations by looking at the steady-state amplitude.

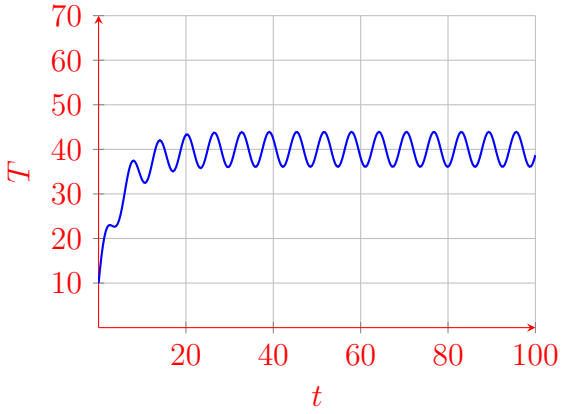
    (a) $\dfrac{dT}{dt} = -2\big(T - (20\cos(t) + 40)\big)$ (c) $\dfrac{dT}{dt} = -0.5\big(T - (20\cos(t) + 40)\big)$

    (b) $\dfrac{dT}{dt} = -1\big(T - (20\cos(t) + 40)\big)$ (d) $\dfrac{dT}{dt} = -0.2\big(T - (20\cos(t) + 40)\big)$

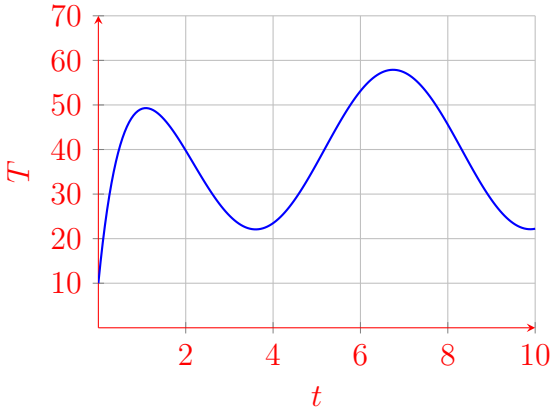
i)



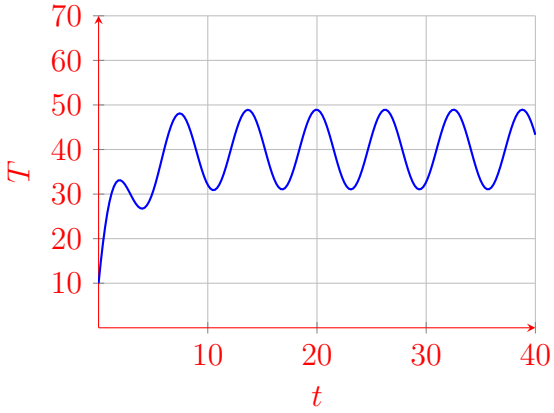
ii)



iii)



iv)



As before, we compare the relative size of k to the final amplitude of the object's temperature. The ambient temperature oscillates with an amplitude of 20 around a baseline of 40.

- **Graph (iii)** tracks the ambient temperature the closest, with the largest eventual amplitude. Thus, it has the largest k-value and matches **(a)** ($k = 2$).
- **Graph (i)** has the next largest amplitude, matching **(b)** ($k = 1$).
- **Graph (iv)** has a fairly small amplitude, matching **(c)** ($k = 0.5$).
- **Graph (ii)** is extremely insulated; after the initial warm-up, it almost flatlines at the average temperature of 40°. This must be the smallest k-value, matching **(d)** ($k = 0.2$).

Mixing

1. A 500 gallon tank is initially filled with pure water. Every minute, 5 gallons of brine containing 0.2 pounds of salt per gallon are pumped into the tank. The well-mixed solution is pumped out at the same rate of 5 gallons per minute. Find a differential equation and initial condition satisfied by $S(t)$, the amount of salt in the tank at time t. Then, solve the initial value problem for $S(t)$.

   Rate In: $\left(0.2\frac{\text{lbs}}{\text{gal}}\right) \times \left(5\frac{\text{gal}}{\text{min}}\right) = 1$ lb/min
   Rate Out: $\left(\frac{S(t)}{500}\frac{\text{lbs}}{\text{gal}}\right) \times \left(5\frac{\text{gal}}{\text{min}}\right) = \frac{S}{100}$ lbs/min

   The differential equation is $\frac{dS}{dt} = 1 - \frac{S}{100}$.
   Because the tank starts with pure water, the initial condition is $S(0) = 0$.

   To solve, we separate variables. First, rewrite the RHS: $\frac{dS}{dt} = -\frac{1}{100}(S - 100)$

   $$\int \frac{1}{S-100}\, dS = \int -\frac{1}{100}\, dt$$

   $$\ln|S - 100| = -\frac{1}{100}t + C$$

   $$S(t) - 100 = Ce^{-t/100} \implies S(t) = Ce^{-t/100} + 100$$

   Using the initial condition $S(0) = 0$:

   $$0 = Ce^0 + 100 \implies C = -100$$

   The specific solution is $S(t) = -100e^{-t/100} + 100$.

2. A 1000 liter tank initially contains 50 kilograms of dissolved salt. Pure water is pumped into the tank at a rate of 10 liters per minute. The well-mixed solution is pumped out at the exact same rate. Find a differential equation and initial condition satisfied by $S(t)$, the amount of salt in the tank at time t. Then, solve the initial value problem for $S(t)$.

   Rate In: Since pure water is flowing in, the rate in is 0 kg/min.
   Rate Out: $\left(\frac{S(t)}{1000}\frac{\text{kg}}{\text{L}}\right) \times \left(10\frac{\text{L}}{\text{min}}\right) = \frac{S}{100}$ kg/min

   The differential equation is $\frac{dS}{dt} = -\frac{S}{100}$.
   The initial condition is $S(0) = 50$.

   To solve, separate variables:

   $$\int \frac{1}{S}\, dS = \int -\frac{1}{100}\, dt$$

   $$\ln|S| = -\frac{1}{100}t + C$$

   $$S(t) = Ce^{-t/100}$$

   Using the initial condition $S(0) = 50$:

   $$50 = Ce^0 \implies C = 50$$

   The specific solution is $S(t) = 50e^{-t/100}$.

3. A 200 gallon tank initially contains 40 pounds of salt dissolved in water. A brine solution containing 0.5 pounds of salt per gallon is pumped into the tank at a rate of 4 gallons per minute. The well-mixed solution is pumped out at a rate of 4 gallons per minute. Find a differential equation and initial condition satisfied by $S(t)$, the amount of salt in the tank at time t. Then, solve the initial value problem for $S(t)$.

Rate In: $\left(0.5\frac{\text{lbs}}{\text{gal}}\right) \times \left(4\frac{\text{gal}}{\text{min}}\right) = 2$ lbs/min

Rate Out: $\left(\frac{S(t)}{200}\frac{\text{lbs}}{\text{gal}}\right) \times \left(4\frac{\text{gal}}{\text{min}}\right) = \frac{S}{50}$ lbs/min

The differential equation is $\frac{dS}{dt} = 2 - \frac{S}{50}$.
The initial condition is $S(0) = 40$.

To solve, we separate variables. Rewrite the RHS: $\frac{dS}{dt} = -\frac{1}{50}(S - 100)$

$$\int \frac{1}{S-100}\, dS = \int -\frac{1}{50}\, dt$$

$$\ln|S - 100| = -\frac{1}{50}t + C$$

$$S(t) = Ce^{-t/50} + 100$$

Using the initial condition $S(0) = 40$:

$$40 = Ce^0 + 100 \implies C = -60$$

The specific solution is $S(t) = -60e^{-t/50} + 100$.

4. A 100 liter tank is initially filled with a solution containing 10 kilograms of a chemical dye. A new solution containing 0.4 kilograms of dye per liter is pumped in at a rate of 2 liters per minute. The well-mixed solution is pumped out at 2 liters per minute. Find a differential equation and initial condition satisfied by $S(t)$, the amount of dye in the tank at time t. Then, solve the initial value problem for $S(t)$.

Rate In: $\left(0.4\frac{\text{kg}}{\text{L}}\right) \times \left(2\frac{\text{L}}{\text{min}}\right) = 0.8$ kg/min

Rate Out: $\left(\frac{S(t)}{100}\frac{\text{kg}}{\text{L}}\right) \times \left(2\frac{\text{L}}{\text{min}}\right) = \frac{S}{50}$ kg/min

The differential equation is $\frac{dS}{dt} = 0.8 - \frac{S}{50}$.
The initial condition is $S(0) = 10$.

To solve, we separate variables. Rewrite the RHS: $\frac{dS}{dt} = -\frac{1}{50}(S - 40)$

$$\int \frac{1}{S-40}\, dS = \int -\frac{1}{50}\, dt$$

$$\ln|S - 40| = -\frac{1}{50}t + C$$

$$S(t) = Ce^{-t/50} + 40$$

Using the initial condition $S(0) = 10$:

$$10 = Ce^0 + 40 \implies C = -30$$

The specific solution is $S(t) = -30e^{-t/50} + 40$.

5. The following four graphs show the amount of salt $S(t)$ (in pounds) in four different tanks over time t (in minutes). All four tanks start with pure water, meaning $S(0) = 0$.

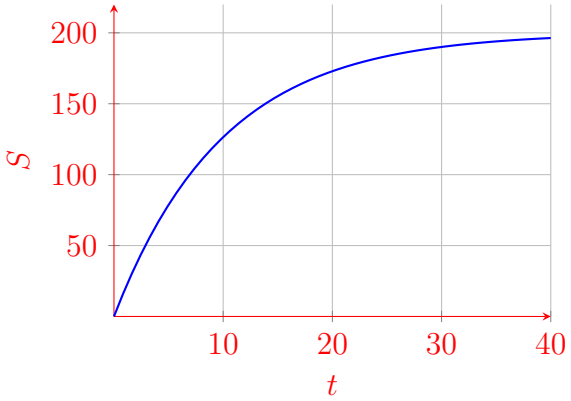
   Match each graph to its corresponding differential equation by considering the **equilibrium value** (which represents the long-term amount of salt, VC) and the **rate of approach** (which is determined by the turnover rate $\frac{R}{V}$).

   Note: Two of the differential equations provided do not match any of the graphs. If an equation has no match, write "None".

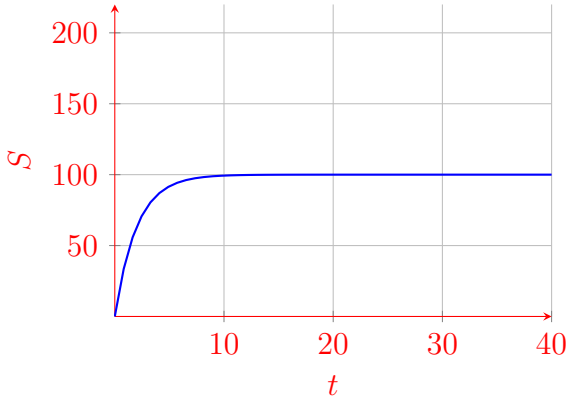
   (a) $\frac{dS}{dt} = 50 - 0.5S$ (c) $\frac{dS}{dt} = 10 - 0.5S$ (e) $\frac{dS}{dt} = 100 - 0.5S$

   (b) $\frac{dS}{dt} = 10 - 0.1S$ (d) $\frac{dS}{dt} = 20 - 0.1S$ (f) $\frac{dS}{dt} = 50 - 0.1S$

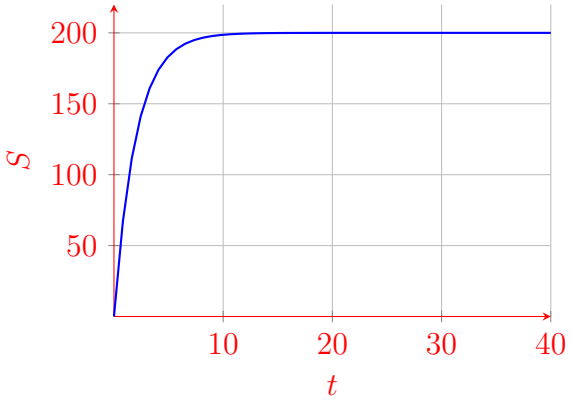
i)



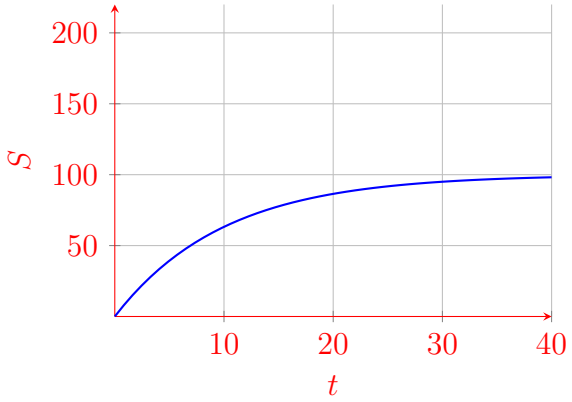
ii)



iii)



iv)



To match the equations, recall that the differential equation for mixing is $\frac{dS}{dt}$ = Rate In $- \frac{R}{V}S$. Factoring out $\frac{R}{V}$ gives $\frac{dS}{dt} = -\frac{R}{V}(S - VC)$. Thus, setting the derivative to zero shows the equilibrium is $S = VC$.

Let's find the equilibrium for each option by setting $\frac{dS}{dt} = 0$: (a) $0 = 50 - 0.5S \implies S = 100$. The "k" value is 0.5. (b) $0 = 10 - 0.1S \implies S = 100$. The "k" value is 0.1. (c) $0 = 10 - 0.5S \implies S = 20$. (d) $0 = 20 - 0.1S \implies S = 200$. The "k" value is 0.1. (e) $0 = 100 - 0.5S \implies S = 200$. The "k" value is 0.5. (f) $0 = 50 - 0.1S \implies S = 500$.

Now look at the graphs:

- **Graphs (i) and (iii)** both level off at an equilibrium of 200. These must be (d) and (e). Graph (iii) approaches 200 much faster than graph (i), so it must correspond to the larger coefficient (0.5).

- **Graphs (ii) and (iv)** both level off at an equilibrium of 100. These must be (a) and (b). Graph (ii) approaches 100 much faster than graph (iv), so it corresponds to the larger coefficient (0.5).

(a) $\frac{dS}{dt} = 50 - 0.5S$ **(ii)** (c) $\frac{dS}{dt} = 10 - 0.5S$ **(None)** (e) $\frac{dS}{dt} = 100 - 0.5S$ **(iii)**
(b) $\frac{dS}{dt} = 10 - 0.1S$ **(iv)** (d) $\frac{dS}{dt} = 20 - 0.1S$ **(i)** (f) $\frac{dS}{dt} = 50 - 0.1S$ **(None)**

6. Match each physical mixing scenario described below with the differential equation that models the amount of salt $S(t)$ in the tank.

   Assume the flow rate in (R) equals the flow rate out (R), so the total volume (V) remains constant. Let C represent the incoming concentration.

   Note: Two of the differential equations provided do not match any of the scenarios. If an equation has no match, write "None".

(a) $\frac{dS}{dt} = 50 - 0.01S$ (c) $\frac{dS}{dt} = 50 - 0.05S$ (e) $\frac{dS}{dt} = 10 - 0.05S$
(b) $\frac{dS}{dt} = 5 - 0.01S$ (d) $\frac{dS}{dt} = -0.05S$ (f) $\frac{dS}{dt} = 5 - 0.1S$

   i) A 1000 L tank. Brine containing 5 kg of salt per liter is pumped in at a rate of 10 L/min.

   ii) A 100 L tank. Brine containing 5 kg of salt per liter is pumped in at a rate of 1 L/min.

   iii) A 1000 L tank. Brine containing 1 kg of salt per liter is pumped in at a rate of 50 L/min.

   iv) A 100 L tank full of brine. **Pure water** is pumped in at a rate of 5 L/min.

   To match these, we calculate the Rate In ($C \times R$) and the coefficient for the Rate Out ($\frac{R}{V}$) for each scenario. The general equation is $\frac{dS}{dt} = CR - \frac{R}{V}S$.

   - **Scenario (i):** $V = 1000$, $R = 10$, $C = 5$. Rate In $= 10 \times 5 = 50$. Rate Out coeff $= \frac{10}{1000} = 0.01$. Equation: $\frac{dS}{dt} = 50 - 0.01S$. This matches **(a)**.
   - **Scenario (ii):** $V = 100$, $R = 1$, $C = 5$. Rate In $= 1 \times 5 = 5$. Rate Out coeff $= \frac{1}{100} = 0.01$. Equation: $\frac{dS}{dt} = 5 - 0.01S$. This matches **(b)**.
   - **Scenario (iii):** $V = 1000$, $R = 50$, $C = 1$. Rate In $= 50 \times 1 = 50$. Rate Out coeff $= \frac{50}{1000} = 0.05$. Equation: $\frac{dS}{dt} = 50 - 0.05S$. This matches **(c)**.
   - **Scenario (iv):** $V = 100$, $R = 5$, $C = 0$ (pure water). Rate In $= 5 \times 0 = 0$. Rate Out coeff $= \frac{5}{100} = 0.05$. Equation: $\frac{dS}{dt} = 0 - 0.05S$. This matches **(d)**.

(a) $\frac{dS}{dt} = 50 - 0.01S$ **(i)** (c) $\frac{dS}{dt} = 50 - 0.05S$ **(iii)** (e) $\frac{dS}{dt} = 10 - 0.05S$ **(None)**
(b) $\frac{dS}{dt} = 5 - 0.01S$ **(ii)** (d) $\frac{dS}{dt} = -0.05S$ **(iv)** (f) $\frac{dS}{dt} = 5 - 0.1S$ **(None)**

7. A 500-gallon tank initially contains 400 gallons of water with 50 lbs of salt dissolved in it. Brine containing 0.2 lbs of salt per gallon is pumped into the tank at a rate of 5 gallons per minute. The well-mixed solution is pumped out *at a faster rate* of 7 gallons per minute. Let $S(t)$ be the amount of salt in the tank at time t in minutes.

(a) Write down (but do not solve) an initial value problem for $S(t)$.

First, find the volume $V(t)$. The tank starts with 400 gallons and loses a net of 2 gallons per minute $(5 - 7 = -2)$.

$$V(t) = 400 - 2t \text{ gallons}$$

Rate In: $\left(0.2\frac{\text{lbs}}{\text{gal}}\right) \times \left(5\frac{\text{gal}}{\text{min}}\right) = 1$ lb/min

Rate Out: $\left(\frac{S(t)}{400-2t}\frac{\text{lbs}}{\text{gal}}\right) \times \left(7\frac{\text{gal}}{\text{min}}\right) = \frac{7S}{400-2t}$ lbs/min

The initial value problem is:

$$\frac{dS}{dt} = 1 - \frac{7S}{400 - 2t}, \quad S(0) = 50$$

(b) What is the domain of this problem? (In other words, how long can this situation continue?)

The situation can only continue as long as there is liquid in the tank. We set the volume equal to zero:

$$400 - 2t = 0 \implies 2t = 400 \implies t = 200 \text{ minutes}$$

The domain is $0 \leq t < 200$.

(c) Does this differential equation have any equilibria? Show your work.

To find an equilibrium, we need to find a constant solution $S(t) = C$ such that $\frac{dS}{dt} = 0$ for all valid t. Setting the derivative to zero:

$$0 = 1 - \frac{7S}{400 - 2t} \implies 1 = \frac{7S}{400 - 2t}$$

$$400 - 2t = 7S \implies S = \frac{400 - 2t}{7}$$

Because S depends on the variable t, there is no constant value of S that makes the derivative zero. Therefore, there are **no equilibria**.

8. A 1000-liter vat initially contains 200 liters of pure water. A liquid dye solution containing 3 kg of dye per liter is pumped into the vat at a rate of 10 liters per minute. The well-mixed solution *is drained out at a slower rate* of 6 liters per minute. Let $S(t)$ be the amount of dye in the vat at time t in minutes.

(a) Write down (but do not solve) an initial value problem for $S(t)$.

First, find the volume $V(t)$. The vat starts with 200 liters and gains a net of 4 liters per minute $(10 - 6 = 4)$.

$$V(t) = 200 + 4t \text{ liters}$$

Rate In: $\left(3\frac{\text{kg}}{\text{L}}\right) \times \left(10\frac{\text{L}}{\text{min}}\right) = 30$ kg/min

Rate Out: $\left(\frac{S(t)}{200+4t}\frac{\text{kg}}{\text{L}}\right) \times \left(6\frac{\text{L}}{\text{min}}\right) = \frac{6S}{200+4t}$ kg/min

Because the vat starts with pure water, the initial amount of dye is 0. The initial value problem is:

$$\frac{dS}{dt} = 30 - \frac{6S}{200 + 4t}, \quad S(0) = 0$$

(b) What is the domain of this problem? (In other words, how long until the vat overflows?)

The situation continues until the vat reaches its maximum capacity of 1000 liters. We set the volume equal to 1000:

$$200 + 4t = 1000 \implies 4t = 800 \implies t = 200 \text{ minutes}$$

The domain is $0 \leq t \leq 200$.

(c) Does this differential equation have any equilibria? Show your work.

To find an equilibrium, we need a constant solution $S(t) = C$ such that $\frac{dS}{dt} = 0$. Setting the derivative to zero:

$$0 = 30 - \frac{6S}{200 + 4t} \implies 30 = \frac{6S}{200 + 4t}$$

$$30(200 + 4t) = 6S \implies 6000 + 120t = 6S$$

$$S = 1000 + 20t$$

Once again, the required value of S changes as time passes. There is no constant value of S that zeros out the derivative, so there are **no equilibria**.

Chemical Rates

1. Consider the following model of a chemical reaction. First, make conjectures regarding how it proceeds, then solve.

$$\begin{aligned} \frac{d[A]}{dt} &= -[A] + 4[B], \quad [A](0) = 5 \\ \frac{d[B]}{dt} &= [A] - 4[B], \quad [B](0) = 0 \end{aligned}$$

What happens to the solutions as $t \to \infty$?

Conjecture: Since $A_0 = 5$ and $B_0 = 0$, the reaction must initially proceed towards B. Because the rate constant for $B \to A$ ($k_2 = 4$) is larger than $A \to B$ ($k_1 = 1$), the system strongly favors A. The final concentration of A should be much higher than B.

Solution: We know that $[A](t) + [B](t) = A_0 + B_0 = 5$. Rearranging gives $[B] = 5 - [A]$. Substitute this into the first differential equation:

$$\frac{d[A]}{dt} = -[A] + 4(5 - [A]) = 20 - 5[A] = -5([A] - 4)$$

Separate variables and integrate:

$$\int \frac{1}{[A] - 4}\, d[A] = \int -5\, dt$$

$$\ln|[A] - 4| = -5t + C \implies [A](t) = 4 + Ce^{-5t}$$

Use the initial condition $A_0 = 5$:

$$5 = 4 + Ce^0 \implies C = 1$$

Thus, $[A](t) = 4 + e^{-5t}$. Since $[B](t) = 5 - [A](t)$, we have $[B](t) = 1 - e^{-5t}$.

As $t \to \infty$, $e^{-5t} \to 0$. Therefore, $[A] \to 4$ and $[B] \to 1$.

2. Consider the following model of a chemical reaction. First, make conjectures regarding how it proceeds, then solve.

$$\begin{aligned} \frac{d[A]}{dt} &= -4[A] + 2[B], \quad [A](0) = 6 \\ \frac{d[B]}{dt} &= 4[A] - 2[B], \quad [B](0) = 0 \end{aligned}$$

What happens to the solutions as $t \to \infty$?

Conjecture: Since we start with only A, it will convert into B. The rate constant for $A \to B$ ($k_1 = 4$) is twice as large as $B \to A$ ($k_2 = 2$), meaning the equilibrium will favor B. The final concentration of B should be twice that of A.

Solution: We know that $[A](t) + [B](t) = A_0 + B_0 = 6$. Rearranging gives $[B] = 6 - [A]$. Substitute this into the first differential equation:

$$\frac{d[A]}{dt} = -4[A] + 2(6 - [A]) = 12 - 6[A] = -6([A] - 2)$$

Separate variables and integrate:

$$\int \frac{1}{[A] - 2}\, d[A] = \int -6\, dt$$

$$\ln|[A]-2| = -6t + C \implies [A](t) = 2 + Ce^{-6t}$$

Use the initial condition $A_0 = 6$:

$$6 = 2 + Ce^0 \implies C = 4$$

Thus, $[A](t) = 2 + 4e^{-6t}$. Since $[B](t) = 6 - [A](t)$, we have $[B](t) = 4 - 4e^{-6t}$.

As $t \to \infty$, $e^{-6t} \to 0$. Therefore, $[A] \to 2$ and $[B] \to 4$.

3. Consider the following model of a chemical reaction. First, make conjectures regarding how it proceeds, then solve.

$$\begin{aligned} \frac{d[A]}{dt} &= -2[A] + 5[B], \quad [A](0) = 0 \\ \frac{d[B]}{dt} &= 2[A] - 5[B], \quad [B](0) = 14 \end{aligned}$$

What happens to the solutions as $t \to \infty$?

Conjecture: We start entirely with B, so B will decrease and A will increase. Because $k_2 = 5$ is significantly larger than $k_1 = 2$, the reaction strongly favors A. The final concentration of A will be much higher than B.

Solution: We know that $[A](t) + [B](t) = A_0 + B_0 = 14$. Rearranging gives $[B] = 14 - [A]$. Substitute this into the first differential equation:

$$\frac{d[A]}{dt} = -2[A] + 5(14 - [A]) = 70 - 7[A] = -7([A] - 10)$$

Separate variables and integrate:

$$\int \frac{1}{[A]-10}\, d[A] = \int -7\, dt$$

$$\ln|[A]-10| = -7t + C \implies [A](t) = 10 + Ce^{-7t}$$

Use the initial condition $A_0 = 0$:

$$0 = 10 + Ce^0 \implies C = -10$$

Thus, $[A](t) = 10 - 10e^{-7t}$. Since $[B](t) = 14 - [A](t)$, we have $[B](t) = 4 + 10e^{-7t}$.

As $t \to \infty$, $e^{-7t} \to 0$. Therefore, $[A] \to 10$ and $[B] \to 4$.

4. Suppose that we have $k_1 = 1$ and $k_2 = 2$, but that the initial concentrations, A_0 and B_0, are not known. Make conjectures as to what the final concentrations of A and B will be, in terms of A_0 and B_0. Now solve the equations above. Your solution will depend on A_0 and B_0. What happens to the solutions as $t \to \infty$?

Conjecture: The rate constant $k_2 = 2$ is twice as large as $k_1 = 1$, which means the reverse reaction is twice as fast as the forward reaction. Thus, at equilibrium, there should be twice as much A as B. The final concentration should be $\frac{2}{3}$ of the total for A, and $\frac{1}{3}$ of the total for B.

Solution: The differential equations are:

$$\frac{d[A]}{dt} = -[A] + 2[B]$$

$$\frac{d[B]}{dt} = [A] - 2[B]$$

We know $[A] + [B] = A_0 + B_0$, so $[B] = (A_0 + B_0) - [A]$. Substitute this into the first equation:

$$\frac{d[A]}{dt} = -[A] + 2(A_0 + B_0 - [A]) = 2(A_0 + B_0) - 3[A] = -3\left([A] - \frac{2}{3}(A_0 + B_0)\right)$$

Separate variables and integrate:

$$\int \frac{1}{[A] - \frac{2}{3}(A_0 + B_0)}\, d[A] = \int -3\, dt$$

$$[A](t) = \frac{2}{3}(A_0 + B_0) + Ce^{-3t}$$

Use the initial condition $[A](0) = A_0$:

$$A_0 = \frac{2}{3}(A_0 + B_0) + C \implies C = \frac{1}{3}A_0 - \frac{2}{3}B_0$$

Thus, the solutions are:

$$[A](t) = \frac{2}{3}(A_0 + B_0) + \left(\frac{1}{3}A_0 - \frac{2}{3}B_0\right)e^{-3t}$$

$$[B](t) = (A_0 + B_0) - [A](t) = \frac{1}{3}(A_0 + B_0) - \left(\frac{1}{3}A_0 - \frac{2}{3}B_0\right)e^{-3t}$$

As $t \to \infty$, $e^{-3t} \to 0$. Therefore, $[A] \to \frac{2}{3}(A_0 + B_0)$ and $[B] \to \frac{1}{3}(A_0 + B_0)$.

5. Suppose that we have $k_1 = 5$ and $k_2 = 3$, but that the initial concentrations, A_0 and B_0, are not known. Make conjectures as to what the final concentrations of A and B will be, in terms of A_0 and B_0. Now solve the equations above. Your solution will depend on A_0 and B_0. What happens to the solutions as $t \to \infty$?

Conjecture: The rate constant $k_1 = 5$ is greater than $k_2 = 3$, so the forward reaction is faster. Therefore, the equilibrium will favor B. The final mixture should be $\frac{3}{8}$ A and $\frac{5}{8}$ B.

Solution: The differential equations are:

$$\frac{d[A]}{dt} = -5[A] + 3[B]$$

$$\frac{d[B]}{dt} = 5[A] - 3[B]$$

Using $[B] = (A_0 + B_0) - [A]$, substitute this into the first equation:

$$\frac{d[A]}{dt} = -5[A] + 3(A_0 + B_0 - [A]) = 3(A_0 + B_0) - 8[A] = -8\left([A] - \frac{3}{8}(A_0 + B_0)\right)$$

Separate variables and integrate:

$$\int \frac{1}{[A] - \frac{3}{8}(A_0 + B_0)}\, d[A] = \int -8\, dt$$

$$[A](t) = \frac{3}{8}(A_0 + B_0) + Ce^{-8t}$$

Use the initial condition $[A](0) = A_0$:

$$A_0 = \frac{3}{8}(A_0 + B_0) + C \implies C = \frac{5}{8}A_0 - \frac{3}{8}B_0$$

Thus, the solutions are:

$$[A](t) = \frac{3}{8}(A_0 + B_0) + \left(\frac{5}{8}A_0 - \frac{3}{8}B_0\right)e^{-8t}$$

$$[B](t) = (A_0 + B_0) - [A](t) = \frac{5}{8}(A_0 + B_0) - \left(\frac{5}{8}A_0 - \frac{3}{8}B_0\right)e^{-8t}$$

As $t \to \infty$, $e^{-8t} \to 0$. Therefore, $[A] \to \frac{3}{8}(A_0 + B_0)$ and $[B] \to \frac{5}{8}(A_0 + B_0)$.

Converting Descriptions to DEs

For each of the following scenarios, translate the verbal description into a differential equation. Be sure to define your variables and use k for any unspecified positive constants of proportionality.

1. A student is memorizing a vocabulary list of 500 words. The rate at which the student learns new words is proportional to the number of words they have not yet learned. Write a differential equation for $W(t)$, the number of words learned at time t.

   The number of words not yet learned is $500 - W$. Because the student is learning (increasing the words known), the rate is positive.

   $$\frac{dW}{dt} = k(500 - W)$$

2. A skydiver falls through the air. The rate of change of their downward velocity is driven by the constant acceleration of gravity, g, but it is reduced by air resistance. The air resistance is proportional to the square of their current velocity. Write a differential equation for the downward velocity $v(t)$.

   The rate of change of velocity is acceleration ($\frac{dv}{dt}$). Gravity increases the downward velocity, while air resistance decreases it.

   $$\frac{dv}{dt} = g - kv^2$$

3. An IV drip delivers a painkiller into a patient's bloodstream at a constant rate of 15 mg per hour. Simultaneously, the patient's liver clears the painkiller from the bloodstream at a rate proportional to the current amount of the drug in the blood. Write a differential equation for $M(t)$, the amount of painkiller in the bloodstream at time t.

   The rate of change is the rate the drug enters minus the rate it is removed.

   $$\frac{dM}{dt} = 15 - kM$$

4. In a town with a population of 25,000, a new rumor is spreading. The rate at which the rumor spreads is proportional to the product of the number of people who have heard it and the number of people who have *not* yet heard it. Write a differential equation for $P(t)$, the number of people who have heard the rumor.

   The number of people who have heard it is P, and the number who haven't is $25,000 - P$.

   $$\frac{dP}{dt} = kP(25,000 - P)$$

5. A retirement account grows with a continuously compounded interest rate of 6% per year. At the same time, the retiree continuously withdraws living expenses at a constant rate of \$45,000 per year. Write a differential equation for $B(t)$, the account balance in dollars at time t in years.

The balance increases by 6% of its current amount, and decreases by the flat withdrawal rate.

$$\frac{dB}{dt} = 0.06B - 45,000$$

6. A spherical raindrop evaporates as it falls. Its volume decreases at a rate proportional to its exposed surface area. Given that the surface area of a sphere is proportional to its volume raised to the two-thirds power ($A \propto V^{2/3}$), write a differential equation for the volume $V(t)$ of the raindrop.

   Since the volume is decreasing, the derivative is negative. We substitute $V^{2/3}$ for the surface area expression.

$$\frac{dV}{dt} = -kV^{2/3}$$

7. The temperature of a hot cup of coffee decreases at a rate proportional to the difference between its own temperature and the constant room temperature of 72°F. Write a differential equation for the temperature of the coffee, $T(t)$.

   Because the coffee is hotter than the room, $T - 72$ is positive. To ensure the temperature decreases, we need a negative sign in front of our positive constant k.

$$\frac{dT}{dt} = -k(T - 72)$$

8. In a certain chemical reaction, reactant A turns into product X. The reaction is "second-order," meaning the rate at which the product X forms is proportional to the square of the amount of reactant A currently remaining. If the initial amount of reactant A was A_0, and each unit of A perfectly converts into one unit of X, write a differential equation for the amount of product $X(t)$.

   Since 1 unit of A makes 1 unit of X, the amount of A remaining at any time is the initial amount minus the amount that has converted $(A_0 - X)$.

$$\frac{dX}{dt} = k(A_0 - X)^2$$